The Commission Paradox: Evidence from Prison Telecommunications*

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Abstract

When an intermediary procures services for end users while receiving a commission from the provider, extracting provider rents can come at the expense of end-user welfare and total surplus. We formalize this *commission paradox* in a first-score auction model and estimate it using records from 37 state-level prison telecommunications procurements. The estimates confirm the paradox: commissions lower provider rents while raising call rates and reducing surplus. A commission ban reduces average rates from $1.37 to $0.58 per 15-minute call and raises total surplus by $4.45 per inmate-month. Competition raises commission bids rather than rate bids; without commissions, competition lowers rates.

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1 Introduction

When an intermediary procures services on behalf of end users, procurement design shapes how surplus is distributed among the provider, the intermediary, and end users (Laffont and Tirole, 1993). A common arrangement is for the provider to pay the intermediary a commission, funded by revenue from end users. Examples include prescription drugs, where pharmacy benefit managers receive rebates from manufacturers (e.g., Ho and Lee, 2026), employer-sponsored retirement plans, where employers select fund providers that make revenue-sharing payments (e.g., Bhattacharya and Illanes, 2025), and grocery retail, where chains receive slotting fees from brand manufacturers (e.g., Hristakeva, 2022). In each case, though price-setting details vary, the intermediary both selects the provider and takes a financial stake in the contract, raising the question of whether procurement mechanisms that serve the intermediary's interest also benefit end users.

We show that, in such settings, procurement designs that more effectively reduce provider rents can also raise end-user prices and reduce total surplus. In standard competitive settings, reducing rents benefits consumers as prices fall toward costs and gains from trade grow. With commissions, the mechanism works differently. The intermediary extracts rents not by demanding lower prices but by claiming a share of revenue, which rewards providers that charge more. Reducing provider rents in this manner shrinks the gains from trade between the provider and end users. Thus, objectives that coincide in standard settings can diverge when the intermediary extracts rents through revenue sharing rather than price competition. We call this the *commission paradox*.

We study the commission paradox in the market for inmate calling services (ICS), where correctional authorities both select the telecommunications provider and receive commission payments. This setting suits our analysis. First, correctional authorities award contracts through scoring auctions in which bids on rates and commissions are combined using predetermined weights, creating a transparent mechanism for tracing how commissions interact with rate-setting. Second, incarcerated individuals and their social contacts, who pay for calls, are external to the procurement process; they neither participate in procurement nor discipline the intermediary by choosing the provider. This absence of consumer choice makes the paradox empirically sharp.

There are more than two million incarcerated individuals in the United States, and aggregate spending on ICS has exceeded $1 billion annually (Shields, 2012). The industry has attracted sustained policy attention, culminating in the FCC's 2024 and 2025 Orders banning commissions and capping rates in telecommunications; the Orders cite prior versions of this paper (FCC, 2024, 2025). As providers have shifted toward tablet-based entertainment and other services not covered by the Orders, correctional authorities face the same challenges in adjacent markets. Our results provide practical guidance on how procurement commissions affect both the rates paid by incarcerated individuals and the revenue flowing to the correctional authority.

The empirical foundation of our study is a set of public records requests submitted to all fifty states regarding ICS procurement for state-level prison systems. We obtained and digitized requests for proposals, winning and losing bids, bid evaluations, and the ensuing contracts. Thus, for a set of procurement events, we have comprehensive data on the entire procurement lifecycle. Observing the full set of submitted bids—including those of losing providers— matters for two reasons. First, it allows us to estimate how procuring entities translate proposed rates and commissions into scores using within-auction variation across bidders, accommodating auction-level heterogeneity in how scores are assigned. Second, losing bids reveal the behavior of higher-cost firms, which is essential for recovering the upper tail of the cost distribution. This matters for counterfactual analysis, where changes in scoring-rule weights can alter which firm wins, making the costs of current losers relevant for predicting outcomes under alternative auction designs.

A salient feature of the data is the relationship between commissions and rates: when commissions enter the scoring rule, they are positively correlated with rates, and rates are substantially higher than in auctions where only rates are subject to bidding. This pattern is consistent with the model's central mechanism, in which commission-based scoring channels competitive pressure toward commission payments rather than rate reductions. Separately, natural experiments in two states that sharply reduced call prices identify demand elasticity, a key model input. New York and New Jersey each implemented large, discrete rate reductions that generated substantial demand responses: call volumes roughly doubled in New York and tripled in New Jersey, while total expenditures fell. These responses inform our welfare calculations.

We model ICS procurement as a first-score auction, following Asker and Cantillon (2008). Each provider privately observes its cost and submits bids comprising a rate and a commission. In equilibrium, each provider's rate maximizes its contract value adjusted for the rate's contribution to the score, while competition enters through the commission. Equilibrium can be characterized in terms of providers' *pseudotypes*, which combine each provider's contract value at its equilibrium rate with the corresponding score adjustment. The expected gap between the best and second-best pseudotype governs provider rents.

Three lemmas characterize how shifting scoring weight toward commissions affects outcomes. Equilibrium rates rise, reducing consumer surplus and total surplus (Lemma 1). At the same time, rents fall because providers become less differentiated. This is because a low-cost provider's advantage is proportional to the number of calls it sells, and higher rates mean fewer calls (Lemmas 2 and 3). Together, the lemmas establish the theorem: greater commission weight extracts provider rents while raising expected rates and lowering expected total surplus (Theorem 1). We also find that increasing the number of bidders raises firms' commission bids, but not their rate bids, so expected rates fall only as the composition of winners improves (Proposition 1).

The separation of rate and commission bidding and the rent-extraction logic follow Che (1993) and Laffont and Tirole (1987), in which a buyer reduces the reward to the non-transfer dimension of a bid to limit information rents. Our setting extends that logic to a case in which the non-transfer dimension is a price paid by end users who are not party to the procurement, so the loss the distortion imposes does not enter the intermediary's revenue. Relative to Che (1993) and Asker and Cantillon (2008), our theoretical contribution is a sharp characterization of this logic and its incidence. Weight on commissions compresses the pseudotype distribution, contracting every expected order-statistic spacing and hence provider rents, while the burden of the higher rates falls on end users; competition, likewise, raises commission payments rather than lowering rate bids.

We then extend the model to allow multidimensional heterogeneity, where providers also differ in quality. The extension introduces selection effects—shifting weight toward commissions alters which firm wins, not only how the winner bids—and provides the foundation for our empirical model of ICS. The paradox cannot be established from primitives alone, as it depends on the joint distribution of costs and quality among competing providers. Quantifying whether it holds in practice and measuring the welfare consequences of procurement design requires taking the model to data. We therefore estimate the structural primitives using the ICS procurement record, recovering the cost and quality distributions that govern equilibrium bidding.

We estimate the model using the simulated method of moments, evaluating the full equilibrium for each candidate parameter vector. The theoretical model features auctions in which commissions are determined through competitive bidding. Our estimation sample also includes auctions in which the procuring entity predetermines commissions. These two auction types require different computational approaches: for the former, we compute equilibrium in closed form following Asker and Cantillon (2008); for the latter, we approximate equilibrium using an iterative best-response algorithm (Bajari, 2001; Armantier et al., 2008). Estimation recovers the scale of within-auction quality heterogeneity and a cost distribution that is consistent with independent estimates from accounting data (Bazelon et al., 2023). The estimated costs fall below the rate cap of $1.35 per 15-minute call described in the FCC's 2025 Order, leaving room for positive margins.

The commission paradox emerges in our quantitative analysis. In numerical simulations, increasing the commission weight in a representative auction compresses the pseudotype distribution, increases competitive pressure, and allows the procuring entity to extract providers' rents through higher commission payments. Equilibrium rates rise in tandem, so consumer surplus and total surplus fall. We find the converse when we implement a commission ban in the auctions in our sample. Averaging across auctions, the ban reduces expected rates from $1.37 to $0.58 per 15-minute call and increases total surplus by $4.45 per inmate-month, as consumer surplus and profit gains more than offset lost commission revenue. Applied to the 1.2 million individuals in state prisons, these estimates imply total surplus gains of about $64 million annually.

We use the estimated model to evaluate additional policy interventions. First, rate regulation that predetermines rates while allowing competition over commissions shifts the cost burden from consumers to the procuring entity, but raises the winning provider's information rents in our simulations: lower regulated rates increase call volume, which widens the cost advantage of low-cost firms and disadvantages their higher-cost rivals in the auction. Second, we examine competition policy. In our simulations, additional bidders compete by raising commission payments rather than lowering rate bids, so expected rates fall only through the composition of winners; banning commissions restores the direct link between competition and rates.

Our paper relates to a growing literature on the role of intermediary commissions in procurement and market design. Commissions appear in the procurement of prescription drugs (Ho and Lee, 2026) and infant formula (Abito et al., 2022; An et al., 2023), as well as in financial and real estate markets where intermediaries receive payments from product providers (Allen et al., 2014; Badoer et al., 2020; Pool et al., 2026; Bhattacharya et al., 2025; Benetton et al., 2025; Grunewald et al., 2023). Among these, Bhattacharya and Illanes (2025) is the closest methodologically, developing and estimating a model of employers' procurement of defined contribution plans to evaluate regulatory efficacy. Relative to this literature, we characterize how commission-based scoring redirects competitive pressure from end-user prices to intermediary payments within an explicit procurement mechanism, and we quantify the resulting division of surplus using the complete bid record of the procurements we study. We apply the framework to public procurement for correctional services, where end users select neither the provider nor the intermediary.

The paper also contributes to the empirical literature on government procurement auctions. Recent work examines how auction design affects outcomes in oil lease auctions (Bhattacharya et al., 2022), federal telecommunications procurement (Kang and Miller, 2022), highway construction (Bolotnyy and Vasserman, 2023), bank resolution (Allen et al., 2024), and defense contracts (Carril et al., 2026). A common feature of this literature is the development of structural models tailored to the institutional details of specific procurement settings. We take the same approach, adapting the scoring auction framework to ICS and using the estimated model to evaluate procurement designs and regulatory interventions relevant to ongoing federal policy.

More broadly, commissions tie the procuring entity's revenue to the prices its providers charge, an arrangement whose incentive properties connect to the literature on delegated decision-making and moral hazard (Grossman and Hart, 1983; Holmström, 1979). We take that arrangement as given and trace its consequence for equilibrium bidding: commissions do not simply transfer surplus, they redirect competitive pressure away from the prices paid by the population the intermediary nominally serves. This points to consumers' ability to select among intermediaries as a key moderating factor; in ICS, incarcerated individuals cannot, so the only discipline on prices is the demand response.[1]

The remainder of the paper proceeds as follows. Section 2 provides institutional background and describes our data. Section 3 develops the theoretical model. Section 4 presents the empirical implementation. Section 5 uses the estimated model to establish the commission paradox in the ICS industry and examine the effects of a commission ban. Section 6 considers additional policy analyses, including rate regulation and competition policy. Section 7 concludes.

[1]The mechanism we examine is distinct from double marginalization (Spengler, 1950), as the intermediary does not set prices to end users; commissions distort equilibrium bidding rather than stacking markups.

2 Inmate Calling Services

2.1 Background

In this section, we describe the ICS industry in greater detail.[2] Three features of the industry are central to our analysis. First, the procuring authority is both the agent that selects the ICS provider and a direct financial beneficiary of the contract through commission payments.[3] Second, incarcerated individuals and their social contacts, who ultimately pay for calls, are excluded from procurement and have no choice of provider. Without outside options, the demand response that would otherwise discipline prices is limited, amplifying the resulting distortion. Third, the scoring auction used to award contracts creates a formal mechanism through which commission payments interact with rate-setting, generating the tradeoffs we model in Section 3. These features are not unique to ICS; similar structures arise in other industries.

Most incarcerated individuals reside in a county jail before sentencing and in a state prison facility after sentencing.[4] Facility-specific rules determine inmates' access to phones and whom they can call, and a number of security measures are in place to ensure compliance. The rules also limit call duration. A typical maximum call length is 15 minutes, but some facilities allow 20- or 30-minute calls. Many calls are collect, so the non-incarcerated recipient pays. Payments from incarcerated individuals and their social contacts are based on pricing schedules set during procurement. Overall call costs can depend on several factors, including call length, whether it is intrastate or interstate, the payment type, and whether there is a fixed connection charge.

The providers are privately owned telecommunications companies. Two providers hold most prison system contracts: Securus and GTL. More providers have contracts with county jails, including IC Solutions and smaller providers like NCIC and CPC. The industry's current configuration reflects a consolidation period in which GTL and Securus acquired many smaller providers. Most recently, GTL acquired Telmate in 2018, Securus abandoned the acquisition of IC Solutions after the Department of Justice expressed antitrust concerns, and IC Solutions acquired CenturyLink's correctional business in 2020.[5] GTL recently rebranded as ViaPath. We are not aware of allegations of collusion in the procurement auctions we study.[6]

[2]We draw on regulatory filings (e.g., FCC, 2012, 2013, 2015, 2021, 2024), an article written by FCC economists (Baker et al., 2020), a small academic literature (e.g., Jackson, 2005; Raher, 2020; Bazelon et al., 2023), and our conversations with industry experts.

[3]For state-level prison systems, the procuring authority is typically a Department of Corrections (DOC). For county jails, it is often a sheriff's office.

[4]The Federal Bureau of Prisons and Immigration and Customs Enforcement also operate facilities that house individuals charged or convicted of violating federal laws and immigration laws, respectively.

[5]The ICS industry began in the 1970s as prison systems relaxed rules restricting incarcerated individuals to a single phone call every three months. AT&T had a monopoly until the 1984 Consent Decree authorized its breakup. Several providers entered the market in the following years, including large telecommunications companies like MCI and Sprint, as well as more specialized providers like GTL. In the 1990s, almost 30 ICS providers competed for prison and jail contracts. For a discussion of this history, see Jackson (2005).

[6]Antitrust litigation in the industry concerns a retail product outside the procurement process: a class action alleged that GTL and Securus fixed the prices of flat-rate single-call services sold to call recipients. GTL settled in 2025; litigation

Correctional authorities use procurement processes that resemble first-price scoring auctions to select their ICS provider. The broad contours are as follows: first, the correctional authority issues a request-for-proposal (RFP) outlining the technical requirements providers must meet, many of which relate to security measures. The RFP also specifies a binding scoring rule describing how the contracting authority will evaluate bids. Second, a formal question-and-answer period allows providers to get additional information about the technical requirements and the facilities involved; prospective bidders can sometimes participate in formal visits ("walk-throughs") of the facilities. Providers also learn about their likely competitors through these interactions. Third, providers submit bids describing their technical capabilities and proposing financial terms. Finally, the correctional authority evaluates the bids according to the scoring rule, and the provider with the highest score wins the contract on the terms it proposes. Contracts typically last three or more years.[7] Figure D.1 provides an illustrative example of the procurement process.

The financial terms in the scoring rule include the pricing schedule and the commission payment. In many procurement processes, the correctional authority predetermines either the pricing schedule or the commission, with the other set through bidding. In other auctions, both are subject to bidding and are usually evaluated separately. Some correctional authorities place the commissions they receive in "inmate welfare" funds that, in principle, support non-essential purchases of books, exercise equipment, and other amenities valued by incarcerated individuals. However, a report of the Prison Policy Initiative, a non-profit advocacy organization, documents that oversight is weak and that funds are sometimes underutilized or misused, for example, to pay for operational expenses that would normally come from the general budget or, in an extreme case, perks for staff (Nam-Sonenstein, 2024). The gap between the stated purpose of commissions and their actual use underscores the potential for misalignment between the procuring authority's financial interest and the welfare of incarcerated individuals.

The large providers have two main revenue streams. The first comes from the prices charged for the phone calls that incarcerated individuals make. We call this "non-fee revenue." The second comes from ancillary fees. Providers obtain these fees in several ways. For example, providers can require calls to be made using prepaid calling cards and levy fees when funds are added to those cards. Typically, commissions are not paid on the revenue obtained from ancillary fees. Fee data are not always shared with the procuring authority and therefore are not always accessible through public record requests; furthermore, some providers set fees that are not state-specific. In our data, we observe fee schedules in some bids (e.g., see Figure D.2) but not others. The available evidence indicates that fee revenue is significant.[8]

against Securus is ongoing. See *Albert v. Global Tel*Link Corp.*, No. 8:20-cv-01936 (D. Md.).

[7]Many procurement processes differ from our description to some extent. For example, some authorities conduct the bidding in two rounds, with a subset of providers being asked to present more information—and possibly better financial terms—in the second round. Furthermore, losing bidders can contest the decision, which occasionally changes the outcome and can lead to an entirely new procurement auction.

[8]The FCC determined that ancillary fees can raise the costs to incarcerated individuals and their social contacts by as much as 40%. See FCC Press Release, October 22, 2015, DOC-335984A1. One provider also recently settled a class

Providing ICS involves at least two major costs. First, when providers start serving clients, they install their own phones and equipment (the telecommunication lines, though, typically do not need to be replaced).[9] Second, providers operate and pay for data centers that store call recordings and associated metadata. Qualitatively, the cost of an "install" increases with the number and size of facilities, and data center costs increase with the calls made. As private equity companies own the large providers, high-fidelity financial information that breaks down the relative magnitude of these costs is not publicly available. However, one study conducts an accounting exercise and estimates the per-minute average call cost to be $0.010-$0.012, depending on the facility's size (Bazelon et al., 2023).

Facilities also incur ICS-related costs. However, these costs appear small. The FCC's 2024 Order accounts for $0.02 per minute in facility costs for prisons (FCC, 2024).

2.2 Regulatory History

The ICS industry has received considerable scrutiny at the federal and state levels. At the federal level, the FCC adopted interim rate caps on interstate calls in 2013 and permanent caps on both intrastate and interstate calls in 2015, but the latter were vacated in a 2017 court decision (*GTL v. FCC*) partly because the FCC lacked statutory authority over intrastate calls. The Martha Wright-Reed Act of 2023 granted that authority and led to the FCC's 2024 and 2025 Orders (FCC, 2024, 2025), which cap rates and ban both commissions and ancillary fees.[10] At least ten states have passed laws eliminating commissions in ICS contracts for state-level facilities, and several have made or scheduled free calls; most of these changes occurred after 2019, when our sample ends.[11] Our counterfactual analyses evaluate relevant policy interventions, including commission bans and rate regulation. We also explore stricter competition policy.

2.3 Data Collection and Summary Statistics

Our data come from public records requests we submitted to all 50 states during the 2020-2021 academic year. Thus, data pertain to state prison systems rather than county jails. We targeted ICS documents and data spanning the previous two decades. In particular, we requested the RFPs, all bids submitted by providers, the bid evaluations and scores, and the contracts with the winning providers. We also requested monthly data on call volumes and the average daily population (ADP)

action lawsuit alleging it seized over $100 million from prepaid accounts following periods of inactivity. See Githieya v. Glob. Tel*Link Corp., USDC (N.D. Ga.), Case No. 1:15-CV-00986.

[9] Incumbents are sometimes thought to have an advantage in the procurement process for this reason; correctional authorities may also perceive them as being more likely to have high-quality technical capabilities.

[10] The 2024 and 2025 Orders cite prior versions of this paper to support demand elasticity estimates and an economic analysis of commissions. The history of federal regulation is detailed in FCC (2024).

[11] The states that eliminated commissions are Colorado, Illinois, Maryland, Nebraska, New Jersey, New Mexico, New York, Ohio, Rhode Island, and South Carolina. States that have implemented or scheduled free calls include Connecticut, California, Colorado, Minnesota, and Massachusetts.

in each prison system.[12] Our empirical analysis is based on the procurement events ("auctions") for which we received the full set of documents (Table D.1). These represent the universe of state-level ICS procurements for which complete records—RFPs, all submitted bids, evaluations, and contracts—were obtainable.

Table 1 shows selected summary statistics at the auction level, on realized financial terms, and the scoring rule weights placed on the rate, the commission, and the providers' technical capabilities. We measure the "rate" as the price of a 15-minute local collect call. For the commission, we use a measure of the payment from the provider per inmate-month that we describe later in this section. We treat "technical capabilities" as a composite of the various considerations that relate to a bidder's ability to meet the buyer's technical requirements.[13] Across all auctions, the average rate is \$1.62, and the average commission payment per inmate-month is \$11.34. The average rate, commission, and technical weights are 0.22, 0.12, and 0.66, respectively. The rate and commission weights can be zero, reflecting that some auctions set them in advance. The average auction has 3.92 bidders.[14]

We allocate the auctions into subsamples based on whether both the rate and the commission are subject to bidding, whether only the commission is subject to bidding, and whether only the rate is subject to bidding. The average rate in these subsamples is \$2.55, \$1.91, and \$0.88, respectively. Two stylized facts are that the rate tends to be lowest in auctions where it is the only financial term subject to bidding, and the commission tends to be largest in auctions where it is the only financial term subject to bidding. We also observe that the rate and commission receive similar weights in the scoring rule when both are subject to bidding, and the combined weight of the financial terms is similar across the subsamples.

Table 2 shows selected summary statistics at the bid level, based on the 155 bids that were submitted across all auctions. We examine the proposed rates and commissions, as well as the assigned scores. The rate and commission statistics are conditional: we restrict the sample for rates and rate scores to bids in auctions that place a positive weight on rates, and analogously for commissions and commission scores. The mean proposed rate is \$1.33, and the mean proposed commission is \$17.52. The scores have means of 0.71, 0.75, and 0.76 for rates, commissions, and technical capabilities, respectively, measured as a share of the maximum possible points.

Figure 1 explores the correlation between rates and commissions in the bid-level data. The left panel uses the full sample of 155 bids; the right panel focuses on the 37 winning bids. Both provide

[12] We submitted public records requests to all 50 states and received at least some information from 43. We obtained a complete set of documents and data on at least one procurement event from 26 states; nine states provided complete records for multiple events. The requests required extensive correspondence over two academic years, as responses depended on each state's compliance requirements, document retention practices, and willingness to engage.

[13] For example, in 2016, North Dakota considered information technology, experience, qualifications, financial strength, and the presentation of each bidder. We treat all of these as part of a bidder's technical capabilities.

[14] All prospective providers submitted a single bid in 33 of the 37 auctions, so the average number of bids (4.19) is close to the average number of bidders. Our empirical model assumes that prospective providers submit a single bid; in estimation, we retain each provider's highest-scoring bid. Allen et al. (2024) explore the conditions under which firms submit multiple bids in scoring auctions.

Table 1: Selected Auction-Level Summary Statistics

	Mean	St. Dev.	10%	25%	50%	75%	90%
All Auctions							
Rate	1.62	1.32	0.27	0.58	1.65	2.25	3.12
Commission	11.34	10.53	0.00	0.05	10.74	19.61	28.27
Rate Weight	0.22	0.23	0.00	0.00	0.20	0.33	0.43
Commission Weight	0.12	0.19	0.00	0.00	0.00	0.15	0.32
Technical Weight	0.66	0.23	0.39	0.60	0.70	0.80	0.88
Number of Bidders	3.92	1.14	3.00	3.00	4.00	5.00	5.00
Rate and Commission in the Scoring Rule							
Rate	2.55	1.98	0.68	0.97	1.65	4.07	4.68
Commission	10.11	7.52	3.15	5.46	10.58	13.18	18.52
Rate Weight	0.19	0.14	0.04	0.10	0.20	0.22	0.32
Commission Weight	0.15	0.18	0.04	0.07	0.10	0.13	0.31
Technical Weight	0.66	0.31	0.40	0.68	0.70	0.80	0.92
Number of Bidders	3.29	0.95	2.60	3.00	3.00	3.50	4.40
Rate Predetermined, Commission in the Scoring Rule							
Rate	1.91	0.39	1.59	1.76	1.90	2.21	2.31
Commission	23.1	8.16	13.99	15.68	24.10	28.57	29.56
Commission Weight	0.33	0.20	0.14	0.20	0.30	0.34	0.48
Technical Weight	0.68	0.20	0.52	0.66	0.70	0.81	0.86
Number of Bidders	4.00	0.67	3.00	4.00	4.00	4.00	5.00
Rate in the Scoring Rule, Commission Predetermined							
Rate	0.88	0.77	0.19	0.33	0.60	1.24	2.06
Commission	4.58	6.34	0.00	0.00	0.62	8.49	13.11
Rate Weight	0.38	0.21	0.20	0.29	0.32	0.41	0.59
Technical Weight	0.62	0.21	0.41	0.59	0.69	0.72	0.80
Number of Bidders	4.17	1.38	2.70	3.00	4.00	5.00	6.00

Notes: The table provides selected summary statistics at the auction level for the 37 distinct auctions in the data. The rate is the cost of a 15-minute local collect call, and the commission is in dollars per inmate-month. Statistics are shown for the full sample and subsamples constructed based on which financial terms are subject to bidding. The full sample includes 37 auctions, and the subsamples include seven, ten, and 18, respectively. Two auctions place no weight on financial terms in the scoring rule and are therefore omitted from the subsamples.

a scatter plot and a line of best fit. A positive empirical relationship between commissions and rates is evident, and we interpret that correlation as a third stylized fact. The slope coefficients in the lines of best fit are statistically significant at the one and ten percent levels, respectively, and the bivariate correlation coefficients are 0.386 and 0.316.

We now return to our measure of commissions. Commission payments are most commonly specified as a percentage of the provider's non-fee revenue. Among auctions with such terms, the average payment is 55% of non-fee revenue. Six auctions result in fixed commission payments that do not depend on revenue. We allocate these payments to the inmate-month level using ADP and contract-duration data. The average such payment is $13.57 per inmate-month. The commission measure reported in Tables 1 and 2 combines percentage and fixed commissions. We first calculate non-fee revenue using rate data and a demand function that yields calls per inmate-month (Section 4.2). We then apply the commission percentage to non-fee revenue and adjust for any fixed commission payments, allocated to the inmate-month level.

Table 2: Selected Bid-Level Summary Statistics

	Mean	St. Dev.	10%	25%	50%	75%	90%
Rate	1.33	1.11	0.27	0.50	0.90	1.73	2.72
Commission	17.52	8.35	5.62	11.35	16.67	23.65	28.28
Rate Score	0.71	0.28	0.32	0.54	0.76	0.99	1.00
Commission Score	0.75	0.25	0.42	0.66	0.83	0.92	1.00
Technical Score	0.76	0.18	0.50	0.64	0.80	0.92	0.98

Notes: The table provides selected summary statistics at the bid level for the 155 bids in the data. The rate is the cost of a 15-minute local collect call, and the commission is in dollars per inmate-month. The scores are the fraction of the total points awarded to the bid. The statistics for rates and commissions are conditional, in that we restrict the sample for rates and rate scores to bids in auctions that place a positive weight on rates ($N = 106$), and analogously for commissions and commission scores ($N = 65$).

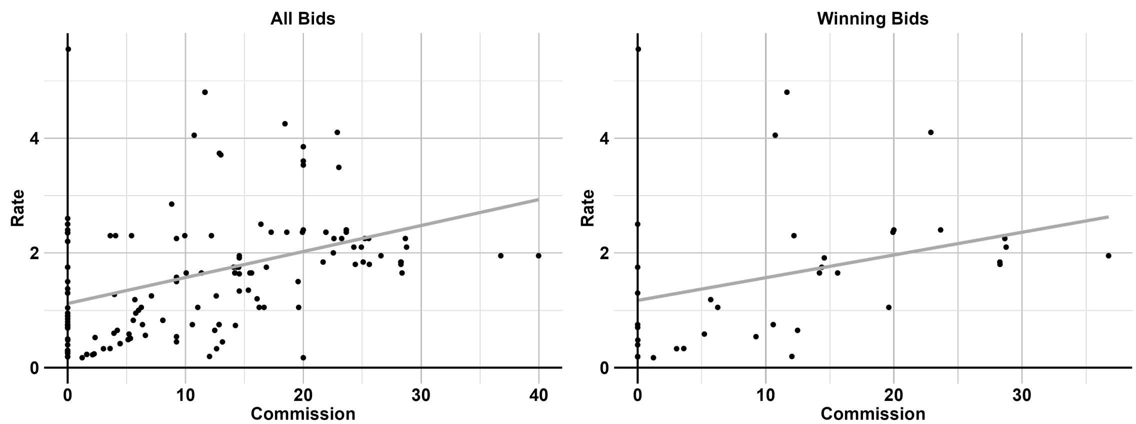

Figure 1: Empirical Relationship Between Rates and Commissions

Notes: The figure provides scatter plots of the commission and the rate in the bid data, using the full sample of 155 bids (left panel) and the 37 winning bids (right panel). We also show lines of best fit. The rate is in dollars per 15-minute local collect call. The commission is in dollars per inmate-month.

3 Theoretical Model

We establish the commission paradox in a procurement model. Commissions intensify competition in the procurement auction and reduce provider rents, but they reduce consumer surplus and total surplus, because commissions are funded by higher prices to end users. We first present a baseline model in which firms differ only in cost, and formally establish the paradox. We then extend the model to multidimensional heterogeneity, and finally consider auctions in which the commission is predetermined.

3.1 Setup and Equilibrium

Procuring entities ("buyers") use sealed-bid first-score auctions to procure a good or service from prospective providers ("firms") on behalf of end users. We index buyers with i and firms with j. Bids comprise the per-unit rate paid by end users, $r \in [\underline{r}, \bar{r}]$, and a commission payment, $k \in \mathbb{R}$, transferred between the buyer and the firm.[15] The buyer specifies a scoring rule, $s_i(\cdot)$, that converts the bids into a single number (a "score"). The firm with the highest score wins the auction at the terms proposed in its bid. The buyers and firms are risk-neutral, and the number of firms that participate in the auction, $J_i \geq 2$, is exogenously determined.

We require the scoring rules to be differentiable, additively separable, and quasi-linear in the commission:

ASSUMPTION 1 (Scoring Rule). *For each auction i, the score is given by*

$$s_i(r, k) = \omega_i^r \, s_i^r(r) + \omega_i^k k,$$

with weights $\omega_i^r \geq 0$ and $\omega_i^k > 0$ ($\omega_i^r + \omega_i^k = 1$). The contribution function, $s_i^r : [\underline{r}, \bar{r}] \to \mathbb{R}$, is continuously differentiable and weakly concave, with $s_i^{r\prime}(r) < 0$.

The concavity of $s^r(\cdot)$ helps ensure a unique equilibrium. Linearity in the commission means a dollar of commission is worth the same number of points regardless of the level, which allows the rate and commission problems to separate. We assume the weights, the contribution function, and the number of bidders are common knowledge. The scoring rules we observe in our data incorporate quality, and some place no weight on the commission; we take up these cases in separate sections.

Each firm knows its cost, c_{ij}, but not its competitors' costs. The cost determines the equilibrium strategy, so it characterizes the firm's type. Formally, we make the following assumption on information and the distribution of firm types:

ASSUMPTION 2 (Information and Distributions). *The cost, c_{ij}, is the private information of firm j, with $c_{ij} \sim^{i.i.d.} F_C$. The distribution F_C is absolutely continuous, with density f_C that is strictly positive on its support, $\mathcal{C} = [\underline{c}, \bar{c}] \subset \mathbb{R}$. The distribution F_C is common knowledge.*

We make the following assumption on contract value, with subscripts denoting derivatives:

ASSUMPTION 3 (Contract Value). *A firm of type c that wins with bid (r, k) earns the contract value, $\Pi(r; c)$, less the commission:*

$$\pi(r, k; c) = \Pi(r; c) - k,$$

with $\Pi \in C^2$ on $[\underline{r}, \bar{r}] \times \mathcal{C}$ and $\Pi_{rr}(r; c) < 0$ throughout. For $r \in [\underline{r}, \bar{r})$, $\Pi_c(r; c) < 0$ and $\Pi_{cr}(r; c) > 0$.

The commission is a lump-sum transfer that does not affect contract value.[16] Contract value is strictly concave in the rate, decreases in cost, and, by the cross-partial condition, is more sensitive to the rate for higher-cost firms.

Auction outcomes also affect end users, who must pay the winning provider's rate to purchase the good or service. Higher rates reduce quantity demanded and consumer surplus. Although

[15] The lower bound $\underline{r}$ accommodates a rate floor set by the buyer or by regulation, including a zero-price floor where rates cannot be negative, and is not otherwise restrictive; the choke rate $\bar{r}$ bounds the domain from above.

[16] This nests commissions specified as a percentage of revenue if the buyer scores the implied dollar payment.

Assumption 3 alone is sufficient to characterize equilibrium (Appendix A.1), incorporating the externality on end users requires demand. We therefore refine our assumption:

ASSUMPTION 4 (Contract Value Refinement). *Contract value takes the form:*

$$\Pi(r; c) = (r - c)q(r)$$

where $q : [\underline{r}, \bar{r}] \to \mathbb{R}_+$*, the demand for calls, is twice continuously differentiable with* $q'(r) < 0$ *for all* $r \in [\underline{r}, \bar{r})$ *and* $q(\bar{r}) = 0$*, where* $\bar{r} < \infty$ *is the choke rate. The cost support satisfies* $\bar{c} < \bar{r}$*.*

The conditions $\Pi_c < 0$ and $\Pi_{cr} > 0$ from Assumption 3 hold under Assumption 4, since $\Pi_c = -q(r)$ and $\Pi_{cr} = -q'(r)$. The condition $\Pi_{rr} < 0$ requires a joint restriction on demand curvature and markups; it holds globally with linear demand, which we use in the empirical model.

The equilibrium concept is symmetric Bayes-Nash equilibrium. An equilibrium strategy is a measurable map $\sigma_i : \mathcal{C} \to [\underline{r}, \bar{r}] \times \mathbb{R}$, with $\sigma_i(c) = \big(r_i(c), k_i(c)\big)$, that maximizes the expected profit of a bidder conditional on other bidders using the same strategy. It satisfies the following condition:

$$\sigma_i(c) = \underset{(r,k)}{\arg\max} \big(\Pi(r; c) - k\big) P_i(r, k) \tag{1}$$

where $P_i(r, k)$ denotes the probability that a firm receives the highest score in auction i when bidding (r, k). Specifically, it equals the probability that $J_i - 1$ competing bidders receive lower scores given i.i.d. costs as in:

$$P_i(r, k) = H_i\big(s_i(r, k)\big)^{J_i - 1} \tag{2}$$

where $H_i(\cdot)$ denotes the cumulative distribution function of each rival bidder's score, which is atomless in equilibrium (Appendix A.1), so ties occur with probability zero.

Finally, we define three notions of surplus. Letting r^* denote the equilibrium rate, these are:

$$\text{Consumer Surplus } (CS) \equiv \int_{r^*}^{\bar{r}} q(p)\, dp$$
$$\text{Total Surplus } (TS) \equiv CS + \Pi$$
$$\text{Public Surplus } (PS) \equiv TS - \pi = CS + k$$

Public surplus accrues to end users and the procuring entity, and excludes provider profit.

3.2 Commissions and Economic Surplus

We now examine how auction design affects outcomes and surplus. To do so, we solve for the equilibrium following Asker and Cantillon (2008). Here, we introduce the two key properties of equilibrium; the full derivation is in Appendix A.1. First, the equilibrium rate of each firm maximizes contract value plus a scoring-rule term that converts the rate's score contribution into dollars:

$$\Pi(r;c) + \psi_i(r) \qquad \text{where} \qquad \psi_i(r) \equiv \frac{\omega_i^r}{\omega_i^k}\, s_i^r(r), \tag{3}$$

so the weights enter only through the ratio ω_i^r/ω_i^k. Equilibrium rate bids, r_{ij}^*, are unique because the objective is strictly concave in the rate under Assumptions 1 and 3, and nondecreasing in cost, because marginal contract value increases in cost. They are also non-strategic because they are independent of the distribution of competing bids. To characterize the downward pressure on rates from the scoring rule, we define the *monopoly rate*, $r_i^M(c) \equiv \arg\max_{r\in[\underline{r},\bar{r}]} \Pi(r;c)$, as the rate that maximizes contract value. Absent a positive weight on rate in the scoring rule, providers' equilibrium bids would feature their monopoly rates.

Second, each bidder's competitive position is determined by its *pseudotype*, defined as the firm's contract value plus the scoring-rule term, both evaluated at the equilibrium rate:

$$x_{ij} = x_i(c_{ij}) \equiv \Pi(r_{ij}^*; c_{ij}) + \psi_i(r_{ij}^*) \tag{4}$$

In equilibrium, the firm with the highest pseudotype wins the auction. We write $X \equiv x(C)$ for the pseudotype of a firm with cost drawn from F_C, and F_X for its distribution. In the second-score version of the auction, the winner's realized profit equals the gap between its pseudotype and the second-best pseudotype, $X^{(1)} - X^{(2)}$, and revenue equivalence ensures that expected profits across the two auction formats coincide conditional on a firm's own type. More generally, the expected gap between the best and second-best pseudotypes governs the winner's expected profit, and the expected commission is expected contract value less that profit.

We impose a regularity condition on the scoring rule that sharpens our formal results. A rule that places heavy weight on rates can drive low-cost firms' rates to the boundary $\underline{r}$, where they no longer respond to the weights, and can drive high-cost firms' rates below cost, where a higher rate raises total surplus. We rule both scenarios out:

ASSUMPTION 5 (Interior Rates and Nonnegative Margins). *The lowest-cost firm's rate is interior, and the highest-cost firm's rate covers its cost:* $r^*(\underline{c}) > \underline{r}$ *and* $r^*(\bar{c}) \geq \bar{c}$.

These conditions place every firm's rate in the interior of $[\underline{r}, \bar{r}]$ and weakly above its cost, as equilibrium rates are nondecreasing in cost.[17]

Our first formal result establishes how the scoring weights affect equilibrium rates and the surplus each provider generates.

Lemma 1 (Comparative Statics). *Under Assumptions 1–5, an increase in the commission weight, with an offsetting decrease in the rate weight, has the following effects at every cost realization:*

(i) *The equilibrium rate increases. Moreover,* $r^*(c) \leq r^M(c)$*, with equality if and only if* $\omega^r = 0$*, and* $r^*(c) \to r^M(c)$ *as* $\omega^r \to 0$.

[17] Appendix A.2 gives the primitive conditions under which these hold and establishes weaker versions of the formal results without Assumption 5.

(ii) *Consumer surplus decreases and contract value increases.*

(iii) *Total surplus decreases.*

All proofs are in Appendix A.2. The rate rises because the scoring penalty on high rates weakens and, simultaneously, the revenue that high rates generate enables providers to offer larger commission payments that matter more in the scoring rule. As the rate weight vanishes, equilibrium rate bids converge to the monopoly rates. Part (ii) follows directly, as higher rates reduce consumer surplus and raise contract value. Part (iii) follows because, with rates at or above cost, higher rates eliminate efficient calls, and the consumer surplus loss exceeds the contract value gain.

Firms differ only in cost, but they compete in pseudotypes. The way cost heterogeneity translates into differences in competitive position depends on the scoring rule. Substituting (3) into (4), the pseudotype is

$$x_i(c) = \frac{\omega_i^r}{\omega_i^k}\, s_i^r(r^*(c)) + \Pi(r^*(c); c) \tag{5}$$

Cost enters both terms indirectly through the rate, and contract value directly. Because the rate is optimal, the indirect effects vanish by the envelope theorem, and the sensitivity of the pseudotype to cost is the direct effect alone. As the contract value is $(r - c)q(r)$, a slight decrease in cost raises the pseudotype in proportion to the number of calls. Formally:

Lemma 2 (Pseudotype Sensitivity). *Under Assumptions 1–4, the pseudotype is strictly decreasing and continuously differentiable in cost, with* $|x'(c)| = q(r^*(c))$ *for all* $c \in \mathcal{C}$.

The weights therefore affect $|x'(c)|$ only through the rate they induce. As the commission weight rises, the equilibrium rate increases and the number of calls falls, so the same cost differences among firms translate into smaller pseudotype differences. This holds at every pair of cost draws:

Lemma 3 (Pseudotype Compression). *Under Assumptions 1–5, an increase in the commission weight, with an offsetting decrease in the rate weight, contracts the pseudotype distribution in the dispersive order: writing* $F_{X'}$ *for the distribution after the increase, for all quantile levels* $0 < \alpha < \beta < 1$,

$$F_{X'}^{-1}(\beta) - F_{X'}^{-1}(\alpha) \;<\; F_X^{-1}(\beta) - F_X^{-1}(\alpha).$$

Consequently, for any fixed number of bidders, $J \geq 2$, *every expected order-statistic spacing is smaller, including the expected gap between the best and second-best pseudotypes.*

A reduction in the dispersive order means that the distance between every pair of quantiles shrinks—the interquartile range, the gap between the 10th and 90th percentiles, and every other spread—so that the distribution tightens everywhere rather than merely on average.[18] The variance of pseudotypes also falls. Weight on commissions moves competition toward a dimension where low-cost firms' advantage is weaker. Lowering rates below the monopoly level sacrifices contract value,

[18] See Section 3.B of Shaked and Shanthikumar (2007) for a formal treatment of the dispersive order.

but less for lower-cost firms, which gain more from the additional calls a lower rate generates, as they have higher price-cost margins.[19] In contrast, a commission reduces profit dollar for dollar regardless of type. As weight shifts to commissions, the differences among firms narrow.

The lemmas combine into the paper's central comparative static:

Theorem 1 (Commission Weight, Rates, and Rents)**.** *Under Assumptions 1–5, an increase in the commission weight, with an offsetting decrease in the rate weight:*

(i) *raises the expected winning rate and lowers expected consumer surplus and expected total surplus;*

(ii) *lowers the expected profit of the winner and raises the expected commission, with the gain in expected commission equal to the loss in expected profit plus the gain in expected contract value.*

Part (i) reflects that the scoring weights do not affect winner identity; by Lemma 2, the pseudotype is strictly decreasing in cost, so the lowest-cost firm wins regardless of the weights, and the pointwise results of Lemma 1 pass directly to expected outcomes. For part (ii), the winner's commission equals contract value minus profit, and both margins move in the buyer's favor: expected profit falls with the compression of the pseudotype distribution (Lemma 3) and contract value rises with the rate (Lemma 1(ii)). The additional commission the buyer receives is thus funded partly by the provider's lost rents and partly by higher rates on end users.

The mechanism connects to the rent-extraction logic of Che (1993) and Laffont and Tirole (1987), in which a buyer can limit information rents by reducing the reward to the non-transfer dimension of a bid. Our setting extends that logic to a case where the non-transfer dimension is a price paid by end users not party to the procurement, creating a negative externality. Thus, mitigating information rents reduces consumer surplus and total surplus. We call this divergence between rent extraction and total surplus the *commission paradox*.

Competition among providers affects commission bids rather than rate bids:

Proposition 1 (Incidence of Competition)**.** *Under Assumptions 1–5 and holding F_C and the scoring rule fixed:*

(i) *The equilibrium rate of every firm is independent of J.*

(ii) *The equilibrium commission bid is weakly increasing in J for every firm and strictly increasing almost surely.*

(iii) *The expected winning rate is strictly decreasing in J, and expected consumer surplus and expected total surplus are strictly increasing in J.*

As the number of bidders changes, each firm adjusts its bid through the commission; its rate is unaffected. Rates fall with competition only through selection, as the winning draw improves.

[19] This is the cross-partial condition $\Pi_{cr} > 0$ imposed in Assumption 3: $\partial^2\Pi/\partial r\, \partial c = -q'(r) > 0$.

The model does not determine whether public surplus rises with the commission weight. Since $PS = TS - \pi$, the change in public surplus is the rent extracted from the winner, net of the loss in total surplus, and the two are governed by the same demand response: the deadweight loss directly, and the rent through the compression of the pseudotype distribution. Neither dominates in general, and the sign depends on demand, the cost distribution, the number of bidders, and the weights. Where public surplus rises, commissions can be redistributed to Pareto-improve the procuring entity and end users.

3.3 Multidimensional Heterogeneity

We now extend the model to incorporate quality as a third bid dimension. Firms differ in both cost and quality. The procuring entity evaluates each firm's quality, v_{ij}^*, and assigns a quality score, $v_{ij} \equiv s_i^v(v_{ij}^*)$, that is a strictly increasing transformation of v_{ij}^*. Each firm knows its quality when it bids, and thus also its quality score. Anticipating the empirical application, we assume that quality does not affect contract value. We replace Assumptions 1 and 2 with the following:

ASSUMPTION 6 (Scoring Rule). *For each auction i, the score is given by*

$$s_i(r,k,v) = \omega_i^r\, s_i^r(r) + \omega_i^k s_i^k(k) + \omega_i^v v,$$

with weights $\omega_i^r, \omega_i^v \geq 0$ and $\omega_i^k > 0$ ($\omega_i^r + \omega_i^k + \omega_i^v = 1$). The contribution functions, $s_i^r(\cdot)$ and $s_i^k(\cdot)$, are twice continuously differentiable with $s_i^{r\prime}(r) < 0$, $s_i^{r\prime\prime}(r) \leq 0$, and $s_i^{k\prime}(k) > 0$. In addition, $s_i^k(\cdot)$ is affine with slope $\alpha > 0$.

ASSUMPTION 7 (Information and Distributions). *The cost, c_{ij}, and the quality score, v_{ij}, are the private information of firm j, with $c_{ij} \sim^{i.i.d.} F_C$ and $v_{ij} \sim^{i.i.d.} F_V$. The distribution F_C is absolutely continuous, with density f_C that is strictly positive on its support, $\mathcal{C} = [\underline{c}, \bar{c}] \subset \mathbb{R}$. The distribution F_V has bounded support, $\mathcal{V} = [\underline{v}, \bar{v}] \subset \mathbb{R}$. Both are common knowledge, and $c_{ij} \perp v_{ij}$ for all ij.*

Under Assumptions 3, 4, 6, and 7, the Bayes-Nash equilibrium features strategies that are contingent on the multidimensional firm type:

$$\sigma_i(c,v) = \underset{(r,k)}{\arg\max}\,\big(\Pi(r;c) - k\big)P_i(r,k;v) \tag{6}$$

where $P_i(r,k;v)$ denotes the probability that a firm with quality score of v receives the highest score in auction i when bidding (r,k). Specifically, it equals the probability that $J_i - 1$ competing bidders receive lower scores given i.i.d. costs and quality scores:

$$P_i(r,k;v) = H_i\big(s_i(r,k,v)\big)^{J_i-1} \tag{7}$$

where $H_i(\cdot)$ denotes the cumulative distribution function of each rival bidder's score. The Asker and Cantillon (2008) characterization of equilibrium extends to this setting (Appendix A.1). In that case, the equilibrium rate again maximizes the sum of contract value and the scoring-rule term, the pseudotype depends on both cost and quality score, and expected profit equals the expected gap between the best and second-best pseudotype. The pseudotype is given by

$$x_i(c,v) = \frac{\omega_i^r}{\omega_i^k}\frac{1}{\alpha}s_i^r(r^*(c)) + \Pi(r^*(c);c) + \frac{\omega_i^v}{\omega_i^k}\frac{1}{\alpha}v \tag{8}$$

Lemma 1 extends under this treatment of multidimensional heterogeneity because the quality term enters the pseudotype additively, with a coefficient that falls as weight shifts toward commissions; indeed, the proof covers this more general setting. Lemma 2 extends to the cost component: $|\partial x/\partial c| = q(r^*(c))$, as the quality term does not depend on cost. The dispersive-order contraction in Lemma 3 is specific to the model without quality, where the pseudotype is a monotone function of cost alone; with quality, it is the sum of independent cost and quality components. Nonetheless, the variance of pseudotypes still falls with the commission weight. By independence,

$$\mathrm{Var}\left(X + \frac{\omega_i^v}{\omega_i^k\alpha}V\right) = \mathrm{Var}(X) + \left(\frac{\omega_i^v}{\omega_i^k\alpha}\right)^2 \mathrm{Var}(V),$$

and both terms fall: the cost component by the argument behind Lemma 3, and the quality component because its coefficient declines as weight shifts toward commissions.[20]

With quality, the lowest-cost firm need not win, so a shift toward commissions alters which firm wins and how each bids. This selection effect depends on the joint distribution of cost and quality, so the commission paradox cannot be established from primitives alone. The direction of selection, however, can be characterized. As the commission weight rises, the pseudotype becomes less sensitive to cost and to quality alike, and the relative decline determines whether selection shifts toward lower-cost or higher-quality firms.

Proposition 2 (Relative Selection). *Under Assumptions 3–7 and $\omega^v > 0$, consider a marginal increase in the commission weight along a differentiable weight path with $d\omega^k > 0$, $d\omega^r \leq 0$, and $d\omega^v \leq 0$. At any c with $r^*(c) \neq 0$, the sensitivity of the pseudotype to cost, relative to its sensitivity to the quality score, increases if and only if*

$$\varepsilon_q(c)\,\varepsilon_r(c) < 1 + \varepsilon_v,$$

where ε_q is the elasticity of demand with respect to the rate, ε_r the elasticity of the equilibrium rate with respect to the commission weight, and ε_v the elasticity of the quality weight with respect to the commission weight, each expressed as a magnitude and evaluated along the path.

The condition has a natural interpretation. From (8), the sensitivity of the pseudotype to the quality score declines mechanically with the commission weight, as it is scaled by ω_i^v/ω_i^k. Cost sensitivity also declines, but through a different channel: the equilibrium rate rises, reducing the

[20] Independence is stronger than needed: the variance result requires only that $E[v \mid c]$ be nonincreasing in cost, so that firm capability drives both lower costs and better quality. It can fail if higher quality raises cost.

number of calls over which cost differences are realized. The product $\varepsilon_q\,\varepsilon_r$ measures the strength of that channel. When it falls below one, cost sensitivity erodes more slowly, and the ranking of firms tilts toward cost. Drawing weight from the quality score further relaxes the bound.

Finally, Proposition 1(i) and (ii) extend, as the separation of the rate and commission problems is unaffected by quality. Competition therefore continues to operate on commissions rather than rates. Part (iii) does not extend: additional bidders can select a higher-cost firm with better quality, so the expected winning rate need not fall.

3.4 Auctions with Predetermined Commissions

Some procuring entities predetermine the commission rather than subject it to bidding. We accommodate this case by relaxing Assumption 6 to permit $\omega_i^k = 0$, holding Assumptions 3, 4, and 7 in place. A firm of type (c, v) facing a predetermined commission $\bar{k}_i$ then chooses only the rate, and the symmetric Bayes-Nash equilibrium strategy solves

$$r_i^*(c, v) = \underset{r \in [\underline{r}, \bar{r}]}{\arg\max} \; \left[\Pi(r; c) - \bar{k}_i\right] P_i(r; v) \tag{9}$$

where $P_i(r; v)$ is the win probability in (7). With the commission fixed, the rate is the only instrument for improving the score. Rate bids are therefore strategic, unlike those in Section 3.2, and depend on the distribution of rivals' costs and quality; no closed-form solution exists. We approximate equilibrium with an iterative best-response algorithm that parameterizes firms' beliefs about rivals' bidding, computes best responses to those beliefs, and updates until the two agree (Bajari, 2001; Armantier et al., 2008). Appendix A.3 describes the algorithm.

4 Empirical Model and Estimation

We introduce an empirical auction model with multidimensional heterogeneity, based on Assumptions 3, 4, 6, and 7, together with the treatment of predetermined commissions in Section 3.4. We estimate the contribution functions, demand function, cost distribution, and the distribution of quality scores. In the data, the quality score is the technical score assigned to each bid in the procurement record, and we use that term hereafter. We adopt parametric specifications for demand and the contribution functions that yield closed-form equilibrium bids, and we place only weak assumptions on the distributions of cost and quality, which determine the pseudotype distribution that governs competitive intensity, the division of contract value between profit and commission, and whether the commission paradox emerges. The cost distribution is nonparametrically identified from observed rates; the distribution of technical scores is identified up to a location-scale structure, with its scale recovered from equilibrium bid moments (Appendix B.1).

4.1 Contribution Functions

We assume the contribution functions are linear in their arguments and known to the bidders:

$$s_i^r(r_{ij}) = \alpha_{0i}^r + \alpha_{1i}^r r_{ij} \tag{10}$$

$$s_i^k(k_{ij}) = \alpha_{0i}^k + \alpha_1^k k_{ij} \tag{11}$$

The commission slope is common across auctions and is the parameter α of Assumption 6: $\alpha = \alpha_1^k$. Thus, we account for unobserved heterogeneity by including fixed effects in the contribution functions. We also allow for limited heterogeneity in the slope parameter in the contribution function for rates, as discussed next. We estimate the parameters using OLS. The main identifying assumption is that the contribution functions are linear (see Appendix B.4 for discussion).

Table 3 shows estimates of the slope parameters. Columns (i)-(iii) focus on rates. Column (i) uses a sample of all bids for which the rate receives a positive weight in the scoring rule; the slope coefficient indicates that if the proposed rate of a 15-minute local collect call is \$1.00 greater, then the associated rate score is 0.184 lower, on average. Columns (ii) and (iii) use subsamples that differ based on whether a positive weight is placed on the commission in the auction's scoring rule (in addition to the rate). The slope coefficient is more negative for auctions that do not consider commissions, and the difference is statistically significant. We use columns (ii) and (iii) for estimation and counterfactuals. Column (iv) shows our commission results. The slope coefficient indicates that a \$1.00 increase in the proposed commission payment per inmate-month is associated with a 0.018 higher commission score, on average. This relationship does not depend significantly on the weight that rates receive in the scoring rule.[21]

4.2 Demand Function

We assume that the number of calls per person is a linear function of the rate. Adding time subscripts, t, to reflect monthly data on calls and rates, the function is

$$q_{it} = \beta_0 + \beta_1 r_{it} + u_{it}^d \tag{12}$$

where q_{it} is the number of calls per inmate-month and u_{it}^d captures seasonal and idiosyncratic factors. We estimate demand using OLS, exploiting variation created by policy-induced price reductions in two states, New York and New Jersey. The identifying assumption is that the rate reductions are orthogonal to changes in the error term. This would be violated if the rate changes

[21] We use our baseline measure of commissions: the payment per inmate-month. We obtain similar results if we use the percentage of revenue to be shared with the procuring entity (and restrict the sample to auctions in which commissions take that form). We omit three bids from the sample that have a commission score of zero. One was in the 2014 Utah auction, and the other two were in the 2019 Utah auction. We suspect the bidders were disqualified for other reasons. If we include these three bids, the slope coefficient increases to 0.025 and remains statistically significant.

Table 3: Estimated Contribution Functions

	Rates			Commissions
	(i)	(ii)	(iii)	(iv)
Rate	-0.184	-0.250	-0.040	
	(0.041)	(0.036)	(0.021)	
Commission				0.018
				(0.009)
R^2	0.925	0.910	0.984	0.972
R^2 (within)	0.213	0.283	0.064	0.155
# of Observations	106	83	23	62

Notes: The table summarizes OLS regression results. Each observation is a bid. The dependent variable is the rate score in columns (i)-(iii) and the commission score in column (iv). We measure the rate score and commission score as the fraction of the maximum available points awarded to a bid. The independent variables are the rate, measured as the price of a 15-minute local collect phone call, and the commission, measured in dollars per inmate-month. All regressions include auction fixed effects. The sample in column (i) includes bids in auctions that place a positive weight on rates. The sample in column (ii) includes bids in auctions that place a positive weight on rates but not on commissions. The sample in column (iii) includes bids in auctions that place a positive weight on both rates and commissions. The sample in column (iv) includes bids in auctions that place a positive weight on commissions. Robust standard errors are in parentheses.

coincided with changes in phone access, but to our knowledge, they did not.[22] We run the regression on the two states separately and together, with and without state-specific constants. Table 4 summarizes the results. We average the coefficients from the state-level regressions to construct the demand function used in the model. The function is: $q = 24.47 - 4.41r$. Appendix C provides a more extensive analysis of these price reductions.

Together, the linear contribution functions and the linear demand function yield closed-form expressions that we use in estimation and counterfactuals. Substituting into the generalized version of (3), the non-commission component of the score, in currency units, becomes

$$\psi_i(r_{ij}, v_{ij}) = \frac{\omega_i^r \alpha_1^r}{\omega_i^k \alpha_1^k} r_{ij} + \frac{\omega_i^v}{\omega_i^k \alpha_1^k} v_{ij} + \frac{\omega_i^k \alpha_{0i}^k + \omega_i^r \alpha_{0i}^r}{\omega_i^k \alpha_1^k} \tag{13}$$

and the equilibrium rates can be expressed:

$$r_{ij}^* = \frac{1}{2}\left(-\frac{\beta_0}{\beta_1} + c_{ij} - \frac{1}{\beta_1} \frac{\omega_i^r \alpha_1^r}{\omega_i^k \alpha_1^k} \right) \tag{14}$$

4.3 Distributions of Costs and Technical Scores

We use simulated method of moments (SMM) to estimate the distributions of costs and technical scores, F_C and F_V, following Pakes and Pollard (1989) and McFadden (1989). The identifying

[22]The midpoint-adjusted arc elasticities are 0.55 and 0.69 for the two policy changes. The FCC's 2025 Order adopts these as its empirical estimates of demand elasticity (FCC, 2025).

Table 4: Demand for Calls

	(i)	(ii)	(iii)	(iv)
Rate	-3.30	-4.37	-4.46	-4.35
	(0.38)	(0.07)	(0.27)	(0.07)
Constant	22.00		19.07	29.86
	(1.35)		(0.61)	(0.29)
NJ Constant		29.90		
		(0.28)		
NY Constant		18.93		
		(0.27)		
Sample	NJ/NY	NJ/NY	NJ	NY

Notes: The table summarizes OLS regression results. Observations are at the state-month level. The dependent variable is the average number of calls per inmate-month. The main independent variable is the price of a 15-minute call. The New York sample includes January–December 2009 and April 2010–March 2011. The New Jersey sample includes January–October 2013 and January–December 2016.

assumption is that differences between observed auction outcomes and those implied by the true distributions are orthogonal to a set of instruments. To assess prediction error, we evaluate equilibrium for each set of candidate distributions. For auctions with endogenous commissions ($\omega_i^k > 0$), we compute the full equilibrium of bids following Asker and Cantillon (2008). For auctions in which commissions are exogenous ($\omega_i^k = 0$), we approximate the full equilibrium using an iterative best-response algorithm following Bajari (2001) and Armantier et al. (2008). Details on these treatments of equilibrium are in Appendices A.1 and A.3, respectively.

For the cost distribution, we use a flexible integrated spline (an I-spline) to parameterize the inverse CDF that maps percentiles to costs. This provides a natural representation because I-splines are continuously differentiable and can be made monotonically non-decreasing under sign constraints, ensuring a valid distribution (see Appendix B.2). For the technical scores, we assume that bidder j in auction i receives

$$v_{ij} = \mu_i + \sigma\, \eta_{ij} \tag{15}$$

where μ_i incorporates auction-level heterogeneity, η_{ij} is an iid shock from a standardized distribution, and σ is a scale parameter that gives the standard deviation of that distribution. We pre-estimate μ_i as the auction-level means of the technical scores using the full sample of 155 bids. This is computationally tractable compared to joint estimation of the auction-specific means alongside the other structural parameters in SMM. [23]

[23] We identify σ from equilibrium information rents, not the raw within-auction variance of technical scores. If we fix σ at the raw within-auction standard deviation of technical scores (see Figure D.4) and re-estimate, technical scores more than account for pseudotype dispersion, the estimated cost distribution collapses to a point mass, and the fit to observed rates and commissions deteriorates. The model is overidentified in this respect, and we resolve the conflict in favor of the bid moments. Appendix B.1 provides details on identification.

Table 5: Results from Simulated Method of Moments

		Estimate	St. Error
Technical Score Scale	σ	0.059	(0.001)
I-Spline Shifter	τ_0	-1.021	(0.077)
I-Spline Weight (First)	τ_1	1.358	(0.025)
I-Spline Weight (Second)	τ_2	0.282	(0.025)
I-Spline Weight (Third)	τ_3	0.118	(0.015)

Notes: The sample comprises all the auctions in our data except Arkansas in 2014 and Michigan in 2018, which place no weight on financial terms, New Hampshire in 2013, which places no weight on technical capability, and Vermont in 2016, which places negligible weight on both financial terms. Sandwich standard errors are in parentheses.

Let $l = 1, \ldots, L$ denote simulated auctions, each comprising draws on costs and technical scores for the auction-specific number of bidders, and $\boldsymbol{\theta}$ be the vector of parameters to be estimated. The empirical moments take the form:

$$\mathbf{m}_r^{(q)}(\boldsymbol{\theta}) = \frac{1}{I}\sum_i \left(r_i^{(q)} - \frac{1}{L}\sum_{l=1}^{L} \tilde{r}_{il}^{(q)}(\boldsymbol{\theta}) \right) \mathbf{Z}_i^r \tag{16}$$

$$\mathbf{m}_k^{(q)}(\boldsymbol{\theta}) = \frac{1}{I}\sum_i \left(k_i^{(q)} - \frac{1}{L}\sum_{l=1}^{L} \tilde{k}_{il}^{(q)}(\boldsymbol{\theta}) \right) \mathbf{Z}_i^k \tag{17}$$

where I is the number of auctions, the superscript (q) identifies the rank of the firms by their overall score, $\tilde{r}_{il}^{(q)}(\boldsymbol{\theta})$ and $\tilde{k}_{il}^{(q)}(\boldsymbol{\theta})$ are the simulated rates and commissions, respectively, and $\mathbf{Z}_i^r$ and $\mathbf{Z}_i^k$ are matrices of instruments. For the instrument matrices, the former includes a constant and the scoring-rule weight on rates, and the latter includes a constant and the scoring-rule weight on commissions. As nearly all auctions feature at least three firms, we construct moments for each of the top three bids. Thus, we have 12 moments in total. Auctions with exogenous commissions contribute only to the rate moments. We use a one-step estimator with $L = 1,000$ and a block-diagonal weighting matrix; each block corresponds to one bid rank and instrument set, with each block retaining only the diagonal elements of $(\mathbf{Z}'\mathbf{Z})^{-1}$, computed separately for $\mathbf{Z}^r$ and $\mathbf{Z}^k$, following Altonji and Segal (1996).

Table 5 contains point estimates and standard errors. The estimated scale of technical scores is 0.059. As the spline parameters are difficult to interpret on their own, we provide the estimated cost distribution in Figure 2, along with a 95% confidence interval. The mean and standard deviation are 0.27 and 0.47, respectively, and the median is 0.48. Costs can be negative, consistent with firms obtaining revenue from ancillary fees. Averaging across auctions and simulation draws, the winning firm's cost is -\$0.07. Thus, our results are consistent with Bazelon et al. (2023), which places explicit costs in the \$0.15-\$0.18 range for a 15-minute call, provided the winning firms obtain some revenue from fees.

For auctions with endogenous commissions, the model implies a mean rate of \$2.10 per 15-

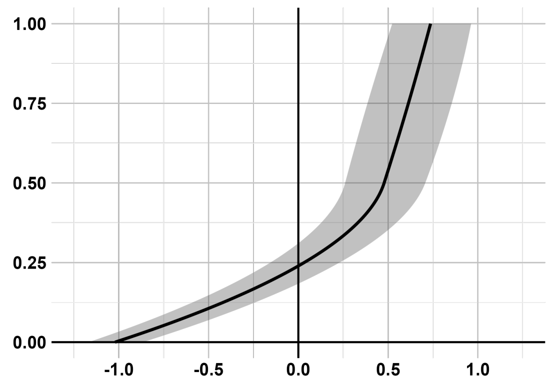

Figure 2: Cumulative Distribution Function of Costs

Notes: The figure plots the CDF of costs (net of ancillary fees) along with a 95% confidence interval. The CDF is obtained by inverting an I-Spline with linear basis functions, a knot at 0.50, and basis function weights that are estimated with simulated method of moments. See Appendix B.2 for details.

minute call and a mean commission of \$17.38 per inmate-month, close to the empirical means of \$2.27 and \$18.54, respectively. The equilibrium rates imply average non-fee revenue of \$31.28 per inmate-month, of which the commission accounts for 56%, leaving the winning firm an average profit of \$14.40 per inmate-month. For auctions with exogenous commissions, the mean model-predicted rate is \$0.71, compared to an empirical mean of \$0.76.

We can also evaluate the condition for relative selection from Proposition 2. Holding the quality weight fixed, auctions select more on cost as the commission weight increases if $\varepsilon_q(c)\,\varepsilon_r(c) < 1$ for all cost realizations. Evaluating the product across all auctions with endogenous commissions and across the support of the cost distribution, the maximum value is 0.29, well below one. Thus, as the commission weight increases, the auctions in our sample tilt toward selecting on cost relative to technical capabilities.

Appendix B.4 examines the sensitivity of these results to the demand specification and the linearity of the contribution functions.

5 Auction Design and the Commission Paradox

We use the empirical model to quantify the commission paradox in the ICS industry. Section 5.1 uses numerical simulations to establish that the paradox emerges under the estimated primitives and to examine how its magnitude varies with market structure and provider heterogeneity. Section 5.2 applies the model to the auctions in our sample, using counterfactual commission bans and maximum commission weights to bracket the welfare consequences of procurement design in ICS.

5.1 Numerical Simulations

We vary the relative weight on commissions and rates in the scoring rule, holding the technical score weight fixed, to trace how expected equilibrium outcomes change as the procurement design shifts toward commissions. Specifically, we set $\omega^v = 0.5$ and let ω^k range from 0.10 to 0.50 with ω^r adjusting one-for-one. We assume four bidders throughout, corresponding to the median number of bidders in the data.

Figure 3 summarizes the results. The top two panels establish the commission paradox. The top-left panel shows that as the commission weight increases, the commission rises, whereas expected consumer surplus and profit both fall. The top-right panel shows that total surplus falls. This is the commission paradox: the auction design that more effectively reduces providers' rents simultaneously decreases total surplus. We also find that the sum of the commission and consumer surplus—what we call public surplus—increases with the commission weight at low values of ω^k, as the commission gains initially offset the consumer surplus losses. These gains, where they exist, accrue entirely to the procuring entity; consumers are strictly worse off. If the procuring entity transfers surplus to incarcerated individuals (e.g., through inmate welfare funds), then modest commission weights may admit Pareto improvements among non-providers. In our simulations, however, the public surplus effects are small.[24]

The bottom-left panel illustrates the mechanism behind these results, showing the empirical density of demeaned pseudotypes under commission weights of 0.10 (solid line) and 0.40 (dashed line). Placing greater weight on the commission reduces providers' rents by compressing the pseudotype distribution, thereby increasing substitutability and intensifying competition (Section 3.2). Indeed, the distribution is visibly tighter with higher commission weight; the corresponding variances are 177.69 and 38.34, and the expected gap between the best and second-best pseudotypes falls from \$13.13 to \$6.23. This compression reflects the auction mechanism alone, as the underlying cost and technical score distributions are held fixed—the pseudotypes simply become less responsive to firm characteristics as the commission weight increases.[25]

The bottom-right panel shows that as the commission weight increases, the winner's expected rate rises (solid line) and expected costs fall (dashed line). The decline in costs is consistent with Proposition 2, the sufficient condition for which we verify in Section 4.3. As the commission weight rises, the pseudotype becomes relatively more sensitive to cost than to the technical score, tilting selection toward lower-cost providers. These lower-cost providers generate higher contract value and can therefore afford larger commissions. Thus, rates increase despite a shift toward low-cost providers.

[24] With enough competition, public surplus decreases with the commission weight because there are fewer rents to extract with the commission, yet rates still rise.

[25] Lemma 3 establishes dispersive-order contraction in the model without multidimensional heterogeneity. The simulations verify the corresponding conclusion in our empirical application with multidimensional heterogeneity: $\mathbb{E}[x^{(1)} - x^{(2)}]$ declines monotonically with the commission weight across the full range we consider.

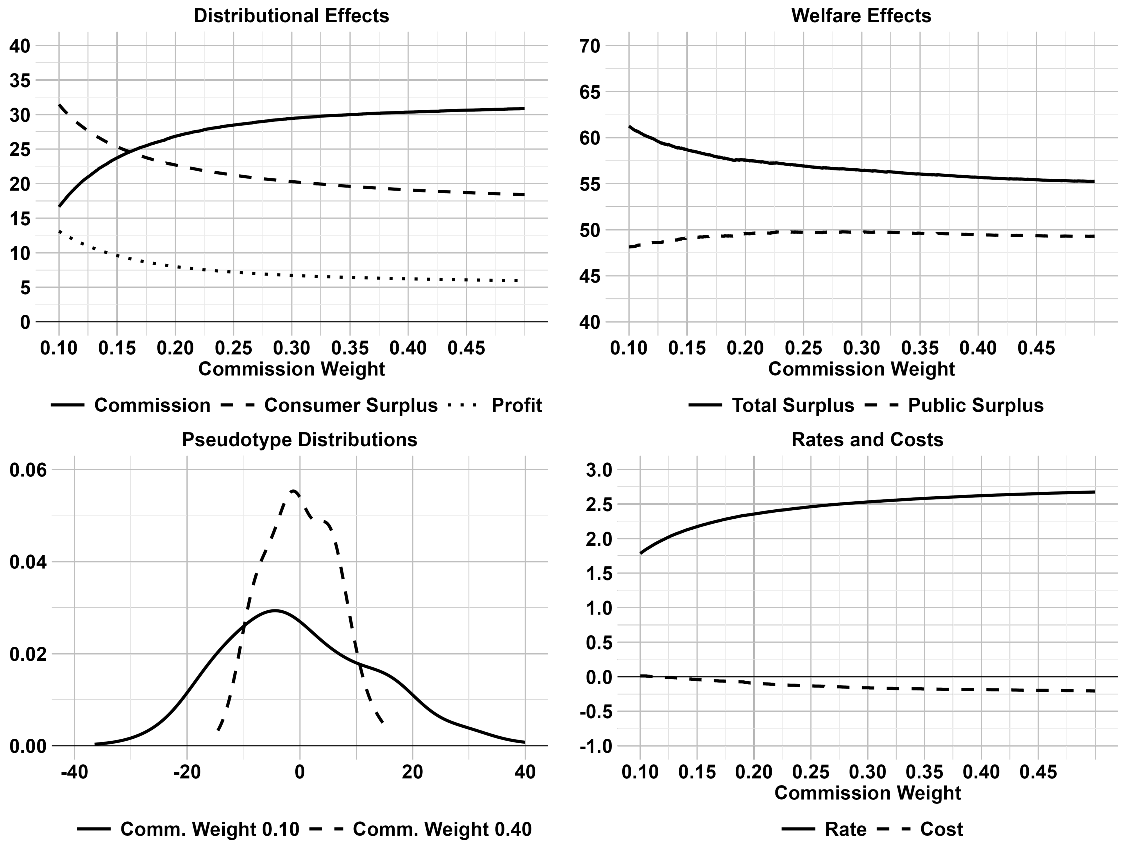

Figure 3: The Commission Paradox

Notes: The panels summarize counterfactual simulations in which the commission weights vary from 0.10 to 0.50 with rate weights shifting one-for-one from 0.40 to 0.00; the technical score weight is 0.50 throughout.

5.2 Commissions in the ICS Industry

Our primary counterfactual is a commission ban. For auctions in which commissions enter the scoring rule, we reallocate the commission weight to rates; for auctions with predetermined commissions, we eliminate the payment.[26] In both cases, we apply the iterative best-response algorithm to obtain equilibrium strategies under the ban. We also simulate a “maximum commission” counterfactual, in which we reallocate all non-quality weight to commissions, to characterize the range of feasible procurement designs. Together, the two counterfactuals capture the extremes of commission policy, from full elimination to maximum weight, given the technical score weights in the sample.

Table 6 summarizes the simulated outcomes before and after the commission ban. Averaging across all auctions, the ban reduces expected rates from \$1.37 to \$0.58 and eliminates commission

[26]With the weight on rates, we use the rate contribution function estimated from auctions without commissions (column (ii) of Table 3), and rates in the ten scored-commission auctions with predetermined rates become subject to competitive bidding.

Table 6: Effects of a Commission Ban in the ICS Industry

	All Auctions			*Scored Commissions*			*Predetermined Commissions*		
	Before	After	Change	Before	After	Change	Before	After	Change
Commission	10.74	0.00	-10.74	17.78	0.00	-17.78	4.12	0.00	-4.12
Consumer Surplus	40.05	54.78	14.74	26.60	53.11	26.51	52.70	56.36	3.66
Profit	13.31	13.76	0.45	14.22	14.57	0.35	12.45	13.00	0.55
Total Surplus	64.09	68.54	4.45	58.60	67.68	9.08	69.27	69.36	0.09
Public Surplus	50.79	54.78	4.00	44.38	53.11	8.73	56.82	56.36	-0.46
Rate	1.37	0.58	-0.79	2.10	0.66	-1.44	0.68	0.50	-0.18

Notes: The table reports simulated outcomes before and after a commission ban, averaged across auctions within each subsample. For auctions with scored commissions, the ban reallocates commission weight to rates and incorporates the steeper rate contribution function from Table 3. For auctions with predetermined commissions, the ban eliminates the predetermined payment. Both "before" and "after" results are based on model simulations. The iterative best-response algorithm is used to approximate equilibrium in all "after" columns and in the "before" period for predetermined commissions. We assume the rate is subject to competitive bidding. The sample comprises 33 auctions: 16 with scored commissions and 17 with predetermined commissions. All values are in dollars per inmate-month except the rate, which is the price of a 15-minute local collect call.

payments of $10.74 per inmate-month. Consumer surplus rises by $14.74 and profit rises by $0.45 per inmate-month, so total surplus increases by $4.45 per inmate-month, as lower rates restore efficient trades that were suppressed in the equilibria with commissions. Public surplus also increases, as consumer gains exceed the lost commission revenue.

The effects are concentrated in auctions where commissions enter the scoring rule. In those auctions, expected rates fall from $2.10 to $0.66, consumer surplus doubles, and total surplus rises by $9.08 per inmate-month. Profit also rises because removing commissions decompresses the pseudotype distribution and widens information rents. This combination—higher profit, lower rates, higher total surplus—confirms the commission paradox. The procuring entity loses $17.78 per inmate-month in commission revenue, but public surplus nonetheless rises by $8.73, as consumer surplus gains more than offset the lost payments. In auctions with predetermined commissions, rates fall from $0.68 to $0.50 and total surplus rises by $0.09 per inmate-month. The smaller magnitudes reflect that predetermined commissions are generally lower, and many are zero—for those auctions, the ban has no effect, and their inclusion in the average attenuates the reported changes.

Our second counterfactual shifts all rate weights to commissions, i.e., sets $\omega_i^k = 1 - \omega_i^v$, and replaces any predetermined commission payments with scored commissions. The rate weight is set to zero; although rates remain subject to competitive bidding, the equilibrium features bids at the monopoly rate. Table 7 reports the results. Averaging across all auctions, rates rise from $1.37 to $2.75 as they move uniformly to the monopoly level, commissions rise from $10.74 to $26.08, and total surplus falls by $11.79 per inmate-month. Public surplus falls by $7.27 per inmate-month.

Together, the two counterfactuals define the range of outcomes available through commission policy. To put these magnitudes in context, the gap in total surplus between the commission ban and the maximum-commission counterfactual is $16.24 per inmate-month. Applied to the approximately 1.2 million individuals in state prisons, and under the assumption that our estimated cost

Table 7: Effects of Maximum Commission Weight in the ICS Industry

	All Auctions			*Scored Commissions*			*Predetermined Commissions*		
	Before	After	Change	Before	After	Change	Before	After	Change
Commission	10.74	26.08	15.34	17.78	24.80	7.03	4.12	27.28	23.17
Consumer Surplus	40.05	17.43	-22.61	26.6	17.29	-9.32	52.70	17.57	-35.13
Profit	13.31	8.79	-4.52	14.22	9.77	-4.45	12.45	7.86	-4.59
Total Surplus	64.09	52.30	-11.79	58.6	51.86	-6.74	69.27	52.72	-16.55
Public Surplus	50.79	43.51	-7.27	44.38	42.09	-2.29	56.82	44.86	-11.96
Rate	1.37	2.75	1.38	2.10	2.77	0.66	0.68	2.74	2.06

Notes: The table reports simulated outcomes, averaged across auctions, before and after maximizing the commission weight. For auctions with scored commissions, the commission weight is set to $1-\omega^v$ and the rate weight to zero. For auctions with predetermined commissions, we replace the predetermined payment with a scored commission weighted by $1-\omega^v$. Both "before" and "after" results are based on model simulations. We use the iterative best-response algorithm to approximate equilibrium in the "before" period with predetermined commissions. We assume the rate is subject to competitive bidding. The sample comprises 33 auctions: 16 with scored commissions and 17 with predetermined commissions. All values are in dollars per inmate-month except the rate, which is the price of a 15-minute local collect call.

and demand parameters are representative across systems, the choice of commission weight alone puts roughly $234 million per year at stake.[27] The status quo captures roughly 73% of these gains ($11.79 of $16.24), primarily due to the auctions already featuring predetermined (often zero) commissions. A commission ban would generate additional total surplus gains of roughly $64 million per year.

6 Additional Policy Analyses

We consider two additional policy analyses: First, we examine a regulation that predetermines rates but allows competition to determine the magnitude of commission payments. Second, to inform competition policy, we examine how outcomes vary with the number of competitors.

6.1 Rate Regulation

We first explore regulation that determines rates; the procuring entity still conducts a scoring auction that determines the commission payment. We examine how outcomes vary with the regulated rate. The analysis can be motivated by recently passed state laws in Connecticut, California, Colorado, Minnesota, and Massachusetts that make calls free for incarcerated individuals; these states pay or will pay for ICS, which could be interpreted as a negative commission.[28] We start with an auction with four bidders and scoring-rule weights of $\omega^k = 0.33$ and $\omega^v = 0.67$. We compute equilibrium for rates from $0.00 to $1.00 per 15-minute call.

[27] County jails, which house an additional 700,000 individuals and often have higher commission rates, are outside our sample but would increase this figure.

[28] For example, Connecticut Senate Bill 972 (2021) states: "The annualized cost for paying the [ICS] vendor for telephone services is approximately $4.5-$5.5 million per year...."

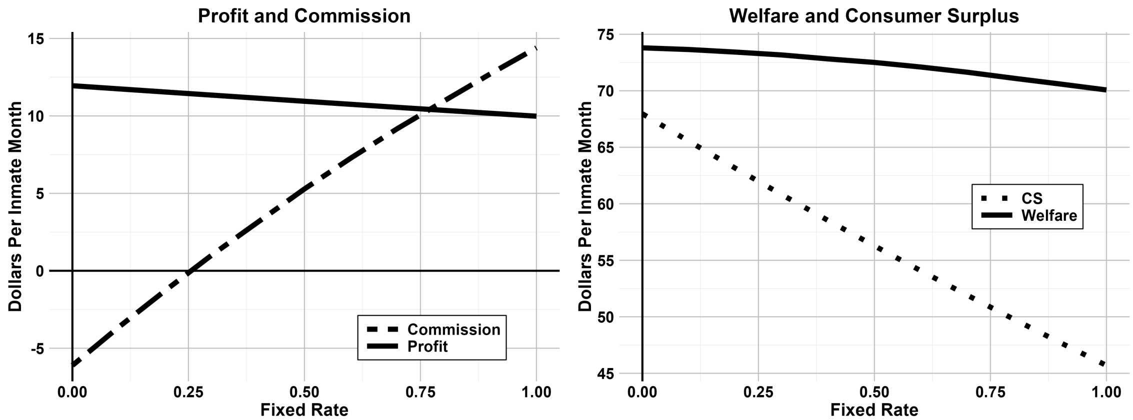

Figure 4: The Economic Effects of Rate Regulation

Notes: The figure is based on 1,000 simulations of an auction with four bidders, scoring rule weights of $\omega^k = 0.33$ and $\omega^v = 0.67$, and regulated rates between \$0.00 and \$1.00 per 15-minute call (horizontal axis). The left panel shows the expected equilibrium commission payment and profit. The right panel shows expected consumer surplus and expected total surplus ($CS + \Pi$).

Figure 4 presents the results. The left panel plots the expected equilibrium commission and profit. With a \$1.00 regulated rate, the commission payment is \$14.40 per inmate-month. As the regulated rate falls (from right to left), the commission decreases because the winning firm earns less revenue from calls and can therefore pay less in commission. For regulated rates below \$0.25, the commission is negative, so the flow of funds reverses and the buyer pays the provider. With free calls, the procuring entity pays the provider \$6.11 per inmate-month. We do not observe payments from the procuring entity to the provider in our data, but they are standard in other settings.

Profit increases as the regulated rate falls. Mechanically, the loss of revenue from rates and fees is more than offset by smaller commission payments. This occurs because rate regulation increases the winning provider's information rent in the procurement auction. At a fixed rate, the pseudotype's sensitivity to cost is $q(r)$, which rises as the regulated rate falls, widening cost-based differentiation among bidders. Although all firms lose profit from calls as regulated rates fall, this effect is amplified for high-cost firms. These disadvantaged firms bid less aggressively, enabling the winning provider to obtain better financial terms. As regulated rates fall, the provider revenue increasingly comes from the buyer via a fixed payment rather than from incarcerated consumers via the rate.

The right panel illustrates the welfare effects of rate regulation. As the regulated rate decreases from right to left, expected consumer surplus under the ICS contract increases (dotted line). Total surplus also increases, but to a more modest extent (solid line). This reflects that, as rates fall, commission payments decline, but this is more than offset by higher consumer surplus and a moderate profit increase. Thus, in the model, rate regulation shifts the cost of ICS from consumers to the procuring entity. Enacted in isolation, it also increases information rents and the profits of the winning firms by disadvantaging higher-cost providers in the auction.

6.2 Competition Policy

Our final analysis examines how outcomes vary with the number of competitors. We motivate the exercise by the industry's extended period of consolidation and the abandoned Securus–IC Solutions merger, which the Department of Justice opposed. In our empirical model, adding a firm affects outcomes for two reasons. First, it reduces the probability that any given firm wins, thereby inducing more aggressive bidding. Second, it improves the winner's expected characteristics (in our model, this implies lower cost and better technical capabilities). We refer to these as a *competition effect* and a *composition effect*, respectively.

We first consider an auction with commissions in the scoring rule. In that context, we can sharpen our description of the composition and competition effects. The composition effect occurs because the expected first-best pseudotype increases with the number of firms, whereas the competition effect occurs because the expected second-best pseudotype increases. We consider an auction with a technical weight of 0.67 and rate and commission weights of 0.165, and compute expected equilibrium outcomes with different numbers of competitors. We report "unadjusted" results that combine the competition and composition effects. We also report "adjusted" results obtained by rescaling the variances of the cost and technical score distributions such that their expected minimums and maximums coincide with those from duopoly. Thus, the expected characteristics of the bidder pool do not vary with the number of firms, approximately isolating the competition effect.

Figure 5 shows the results. In the top panels, both the rate and the commission are subject to competitive bidding. The top-left panel focuses on the expected equilibrium rate. As we increase the number of firms from two to ten, the unadjusted rate decreases from $2.51 to $2.22, whereas the adjusted rate remains flat. Thus, any changes in the rate reflect compositional effects. This tracks our theoretical results, which establish that the winning firm chooses rates to maximize contract value plus the scoring-rule term, independent of the identity or number of competitors. The top-right panel, in contrast, shows that the expected equilibrium commission increases with the number of firms, in both the unadjusted and adjusted cases. These results indicate that adding more firms benefits the procuring entity when commissions are subject to competitive bidding, but whether consumers gain depends on the characteristics of the new firms.

The bottom panel considers auctions in which commissions are not in the scoring rule. In that setting, the rate is the only financial term subject to competition, and adding firms lowers rates in both the adjusted and unadjusted exercises. In our experiments, we assume a technical weight of 0.67 and a rate weight of 0.33. As shown, both the unadjusted and adjusted expected equilibrium rates decrease. Unadjusted rates decline from $1.23 to about -$0.42, whereas adjusted rates decrease to $0.11 as we increase the number of firms from two to ten. (Recall that costs in our model are net of fees and so can be negative.) When commissions are absent, the link between competition and rates is robust.

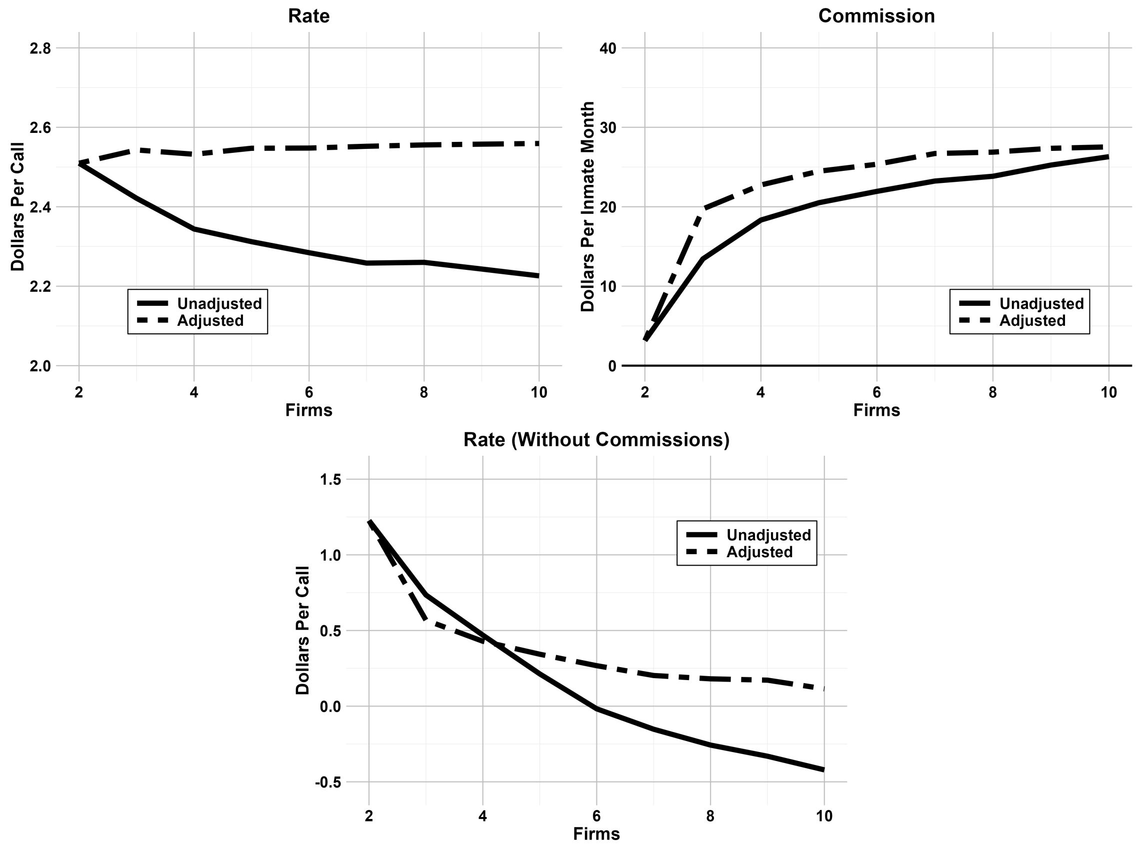

Figure 5: The Effect of Competition on Rates and Commissions

Notes: The panels depict simulated rates and commissions as the number of firms varies from two to ten. In the top panels, both the rate and the commission are subject to competitive bidding. In the bottom panel, only the rate is subject to competitive bidding; there is no commission. The unadjusted trends combine two effects: a competition effect (more aggressive bidding) and a composition effect (more firms lower the expected cost of the winner). Adjusted trends rescale the cost and technical score distributions so that the expected best cost and quality remain constant, isolating the competition effect.

7 Conclusion

Procurement designs that more effectively reduce provider rents do not necessarily improve end-user outcomes. In the ICS industry, we find that shifting scoring weight toward commissions reduces provider rents, but raises rates and reduces total surplus—the commission paradox. We connect this to economic primitives using a first-score auction model in the spirit of Asker and Cantillon (2008). As scoring weight shifts toward commissions, providers internalize less downward pressure on rates, and equilibrium rates rise toward monopoly levels. At the same time, providers' competitive positions become less differentiated, eroding rents. These two forces—higher rates and lower rents— stem from the same shift in the scoring rule. Thus, a primary tool intermediaries can use to extract upstream rents also can harm downstream consumers.

The trade-off we identify between rent extraction through commissions and end-user welfare may be relevant in other intermediated procurement settings, though institutional differences will determine how strongly it operates. A key determinant is likely consumers' ability to select among intermediaries. When intermediaries compete for consumers, competition may discipline commissions and attenuate the paradox. However, if commissions can be shared with consumers—through lower fees, better service, or direct transfers—competition could instead sustain or amplify commissions in equilibrium. In this light, ICS represents one pole: incarcerated individuals cannot switch providers and receive no such transfers, so the only discipline on prices is the demand response. Whether competition among intermediaries could attenuate the paradox in other settings remains an empirical question.

Within ICS specifically, recent state-level commission bans and the FCC's 2024 and 2025 Orders create opportunities to test the commission paradox with reduced-form methods, complementing our model-based approach. More broadly, Raher (2020) highlights that prices are inflated for incarcerated individuals across many goods and services beyond telecommunications, suggesting that procurement distortions in correctional settings could extend well beyond ICS. Future research could examine how call prices affect broader outcomes such as recidivism and post-release employment, or explore the effects of visitation and social connectedness more broadly.

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Appendix

For Online Publication

A Theoretical Model

We provide details on the Bayes-Nash equilibrium in auctions with commissions in the scoring rule ($\omega_i^k > 0$), prove the results in Section 3, and finally provide more details about how we approximate Bayes-Nash equilibrium for other auctions ($\omega_i^k = 0$).

A.1 Equilibrium with Commissions in the Scoring Rule

We characterize equilibrium for auctions in which commissions enter the scoring rule ($\omega_i^k > 0$) based on a reformulation of the model developed in Asker and Cantillon (2008). We use Assumptions 6 and 7, which extend Assumptions 1 and 2 to quality and to a general commission slope, together with Assumptions 3 and 4; the model of Section 3.2 is the case $\omega_i^v = 0$, $\alpha = 1$, and $s_i^k(k) = k$. Throughout, $J_i \geq 2$, rates lie in $[\underline{r}, \bar{r}]$, and commissions are unrestricted in sign and magnitude.

Reduction to a One-Dimensional Auction

Dividing the score by $\omega_i^k \alpha$ yields a normalized score that ranks bids identically to s_i and is measured in dollars:

$$z \equiv \frac{s_i(r,k,v)}{\omega_i^k \alpha} = k + \psi_i(r,v), \qquad \psi_i(r,v) \equiv \frac{\omega_i^r}{\omega_i^k \alpha}\, s_i^r(r) + \frac{\omega_i^v}{\omega_i^k \alpha}\, v + \frac{s_i^k(0)}{\alpha} \tag{A.1}$$

where the last term absorbs any bidder-invariant intercept in $s_i^k(\cdot)$ and vanishes under Assumption 1. A bid (r, k) is equivalent to a pair (r, z), with the commission recovered as

$$k = z - \psi_i(r,v) \tag{A.2}$$

Substituting into (6), expected profit is

$$\big[\Pi(r;c) + \psi_i(r,v) - z\big]\, P_i(z) \tag{A.3}$$

where $P_i(z)$ is the probability that z exceeds the normalized scores of the $J_i - 1$ rivals. The change of variables is global rather than local. Because the commission is unrestricted, every rate is attainable at any target z, so the firm's problem separates: for any z with $P_i(z) > 0$, the rate uniquely maximizes the bracketed term, so that

$$r_i^*(c) = \arg\max_{r \in [\underline{r}, \bar{r}]} \big\{\Pi(r; c) + \kappa_i \, s_i^r(r)\big\}, \qquad \kappa_i \equiv \frac{\omega_i^r}{\omega_i^k \alpha} \tag{A.4}$$

independent of z, of rivals' strategies, and—because ψ_i is additively separable—of the quality score. A firm whose target wins with probability zero is indifferent over rates; in equilibrium only the lowest pseudotype is in this position, and we assign it the rate in (A.4) by continuity, which affects no equilibrium outcome. The maximizer is unique, because $\Pi(\cdot; c)$ is strictly concave and s_i^r is concave, and it is nondecreasing in c, because $\Pi_{cr} > 0$ (Facts 1 and 3 in Appendix A.2). Thus, equilibrium rate bids are non-strategic. This generalizes (3) to the multidimensional model.

Let $x_i(c, v) \equiv \max_r \{\Pi(r; c) + \psi_i(r, v)\}$ denote the value of the rate problem, the firm's *pseudotype* defined in (4) and, with quality, in (8), and let $b_i(x)$ denote the normalized score bid of a firm with pseudotype x. The remaining problem,

$$b_i\big(x_i(c, v)\big) = \arg\max_{z} \ \big(x_i(c, v) - z\big) \, P_i(z) \tag{A.5}$$

is a symmetric independent-private-values first-price auction in which each firm's valuation is its pseudotype. Firms with the same pseudotype adopt the same strategy, so the multidimensional type space (c, v) collapses to a single index, as in Asker and Cantillon (2008), and the bidding strategy in (6) is $\sigma_i(c, v) = \big(r_i^*(c), \, b_i(x_i(c, v)) - \psi_i(r_i^*(c), v)\big)$ by (A.2). The score bid $b_i(\cdot)$ is strictly increasing, and the firm with the highest pseudotype wins.

First-Order Conditions

Under Assumption 5, equilibrium rates are interior and satisfy

$$\Pi_r(r; c) + \kappa_i \, s_i^{r\prime}(r) = 0 \tag{A.6}$$

which is necessary and sufficient given strict concavity of the objective in (A.4); rates are then strictly increasing in cost (Appendix A.2). The condition coincides with the first-order conditions of the original problem (6). For bids with $P_i > 0$, differentiating expected profit with respect to the rate and the commission, and rearranging, yields

$$\Pi_r(r; c) = -\frac{P_r}{P_k}(r, k; v) \tag{A.7}$$

$$\Pi(r; c) = k + \frac{P}{P_k}(r, k; v) \tag{A.8}$$

where subscripts denote derivatives and $P_i(r, k; v)$ is the win probability in (7). The first-order condition in (A.7) equates the marginal profit from raising the rate to the marginal rate of substitution between rate score and commission score in the scoring rule. The first-order condition in (A.8) determines how aggressively the firm bids on commissions, weighing the cost of higher payments against an improved win probability.[29] Under Assumption 6, the iso-probability slope is

$$\frac{P_r}{P_k}(r,k;v) = \frac{P_i'(s)\,\omega_i^r s_i^{r\prime}(r)}{P_i'(s)\,\omega_i^k \alpha} = \kappa_i\, s_i^{r\prime}(r) \tag{A.9}$$

independent of v by additivity and of k by linearity of $s_i^k(\cdot)$, so (A.7) is (A.6). The first-order conditions presuppose a differentiable win probability and an interior rate; the reduction above requires neither, which is why we use it to establish the equilibrium and reserve the first-order conditions for identification (Appendix B.1).

Pseudotypes and Equilibrium Commissions

The win probability is a function of the distribution of pseudotypes (denoted by $T_i(\cdot)$) and equals the probability that $J_i - 1$ bidders have a lower pseudotype:

$$P_i(z) = T_i\big(b_i^{-1}(z)\big)^{J_i-1} \tag{A.10}$$

The distribution of pseudotypes is given by

$$T_i(x) = \iint \mathbb{1}\big(x_i^*(c,v) \le x\big)\, dF_C(c)\, dF_V(v) \tag{A.11}$$

where $x_i^*(c,v)$ is the pseudotype of a bidder with cost c and quality score v. The distribution is atomless. Without quality, the pseudotype is a continuous, strictly decreasing function of an atomless cost (Lemma 2); with quality, the cost component retains that property, and the sum, which includes an independent absolutely continuous term, has a density. We can now derive a closed-form solution. Denoting the support of $T_i(\cdot)$ by $[\underline{t}, \bar{t}]$, the equilibrium normalized score bid is

$$z_{ij}^* = b_i(x_{ij}^*) = x_{ij}^* - \frac{\int_{\underline{t}}^{x_{ij}^*} T_i(u)^{J_i-1} du}{T_i(x_{ij}^*)^{J_i-1}} \qquad \text{for } x_{ij}^* \in (\underline{t}, \bar{t}] \tag{A.12}$$

with the boundary condition $b_i(\underline{t}) = \underline{t}$, to which the expression converges as $x_{ij}^* \downarrow \underline{t}$. Finally, the equilibrium commission is given by

$$k_{ij}^* = z_{ij}^* - \psi_i(r_{ij}^*, v_{ij}) \tag{A.13}$$

[29] Analogous first-order conditions obtain generally in scoring auction models and can form the basis of nonparametric identification and estimation under a best-response approach (Guerre et al., 2000; Kong et al., 2022). The first-order condition in (A.7) is obtained by plugging (A.8) into the derivative of expected profit with respect to the rate. It thus characterizes profit-maximizing rates conditional on the firm bidding a profit-maximizing commission.

Equations (A.4) and (A.13) characterize the symmetric Bayes-Nash equilibrium for any auction in which commissions receive weight in the scoring rule; it is unique up to the bid of the lowest pseudotype, which wins with probability zero (Che, 1993; Asker and Cantillon, 2008). If rates receive no weight, ψ_i does not depend on the rate, every firm bids its monopoly rate, and the auction is an independent private values first-price auction in commissions for which (A.13) alone characterizes the equilibrium.

For estimation, we compute the full equilibrium for each candidate parameter vector by evaluating (A.12) directly, estimating $T_i(\cdot)$ with a Gaussian kernel with a plug-in bandwidth, and computing the integral via cumulative trapezoidal integration on a fine grid. Equilibrium commissions then follow from (A.13). In many of our counterfactuals, for which we need only the commission of the winning bidder, we follow Laffont et al. (1995) and use revenue equivalence to circumvent solving the full equilibrium. Revenue equivalence holds in our setting under quasi-linearity, risk neutrality, and i.i.d. types (Myerson, 1981). As the allocation depends only on pseudotypes, equilibrium payments can be characterized using a second-score construction: the winner's normalized score equals the second-highest pseudotype among bidders. From (A.2), this determines the commission paid by the winner. In this construction, the profit of the winner equals $X^{(1)} - X^{(2)}$, where the superscripts refer to the rank of pseudotypes. The construction reproduces expected commissions and profits. Realized first-score outcomes differ state by state, so statistics beyond means require the full equilibrium, which we use in estimation.

A.2 Proofs

Preliminaries

We prove the results of Sections 3.2 and 3.3 under Assumptions 3, 4, 6, 7, and 5, using the notation of the multidimensional model; the model of Section 3.2 is the case $\omega^v = 0$ and $\alpha = 1$. Facts 1–4, Lemma 1, Lemma 2, and Proposition 1(i)–(ii) hold in the multidimensional model: the first three concern the rate problem and the cost component of the pseudotype, neither of which involves the quality score, and the proof of Proposition 1(ii) uses only atomlessness of the pseudotype distribution. Lemma 3, Theorem 1, and Proposition 1(iii) are proved for the model without quality, $\omega^v = 0$, where the pseudotype is a strictly monotone function of cost alone: with multidimensional heterogeneity, the pseudotype distribution need not contract in the dispersive order and the lowest-cost bidder need not win (Section 3.3). Let $\kappa \equiv \omega^r/(\omega^k \alpha)$ and $a \equiv \omega^v/(\omega^k \alpha)$. By (A.4), the equilibrium rate solves

$$r^*(c;\kappa) = \arg\max_{r\in[\underline{r},\bar{r}]} g(r,c;\kappa), \qquad g(r,c;\kappa) \equiv (r-c)\,q(r) + \kappa\, s^r(r)$$

and we write $x(c;\kappa) \equiv \max_r g(r,c;\kappa)$ for the cost component of the pseudotype, so that the pseudotype is $x(c;\kappa) + a\,v$ in the multidimensional model and $x(c;\kappa)$ in the model of Section 3.2. An increase in the commission weight, with an offsetting decrease in the rate weight, is a comparison of two configurations with $\kappa_1 < \kappa_0$; in the multidimensional model, where weight may also be drawn from the quality score, $a_1 \leq a_0$ as well, and $\kappa_1 < \kappa_0$ holds whenever $\omega_0^r > 0$. The case $\kappa_1 = 0$ is permitted. We write $r_i^*(c) \equiv r^*(c; \kappa_i)$, $x_i(c) \equiv x(c; \kappa_i)$, and $X_i \equiv x_i(C)$ for $C \sim F_C$, with $i \in \{0, 1\}$ indexing the two configurations. Assumption 5 is imposed at $i = 0$.

Facts 1–3 are used throughout and do not use Assumption 5; Fact 4 records the implication of Assumption 5 used in the total surplus comparisons.

Fact 1 (Existence, uniqueness, continuity). For each (c, κ), $g(\cdot, c; \kappa)$ is continuous on the compact set $[\underline{r}, \bar{r}]$ and strictly concave, because $\Pi_{rr} < 0$ (Assumption 3) and $s^{r\prime\prime} \leq 0$ (Assumption 6). Hence $r^*(c; \kappa)$ exists, is unique, and, by the theorem of the maximum, is continuous in (c, κ).

Fact 2 (Rates below the choke rate). For any $c \in \mathcal{C}$ and $\kappa \geq 0$, $g(\bar{r}, c; \kappa) = \kappa s^r(\bar{r}) < g(r', c; \kappa)$ for r' slightly below $\bar{r}$: contract value $(r' - c)q(r')$ is strictly positive for $r' \in (\max\{c, \underline{r}\}, \bar{r})$, which is nonempty because $\bar{c} < \bar{r}$ (Assumption 4), while $\kappa s^r(r') \geq \kappa s^r(\bar{r})$ because s^r is decreasing. Hence $r^*(c; \kappa) < \bar{r}$ and $q(r^*(c; \kappa)) > 0$.

Fact 3 (Monotonicity in cost). For $c_1 > c_0$ and any κ, revealed preference at the two optima gives

$$g(r_1, c_1; \kappa) \geq g(r_0, c_1; \kappa) \quad \text{and} \quad g(r_0, c_0; \kappa) \geq g(r_1, c_0; \kappa),$$

where $r_m \equiv r^*(c_m; \kappa)$. Adding the two inequalities and cancelling the κs^r terms yields $\int_{c_0}^{c_1} \big[\Pi_c(r_1; u) - \Pi_c(r_0; u)\big]\, du \geq 0$. Since $\Pi_{cr} > 0$ on $[\underline{r}, \bar{r})$ and both rates lie there by Fact 2, the integrand is strictly negative throughout if $r_1 < r_0$, so $r_1 \geq r_0$: equilibrium rates are nondecreasing in cost. Where rates are interior they are strictly increasing, because the first-order condition $q(r) + (r - c)q'(r) + \kappa s^{r\prime}(r) = 0$ cannot hold at the same r for two different costs.

Fact 4 (Nonnegative margins). Suppose Assumption 5 holds at configuration κ. Then $r^*(c; \kappa) \geq c$ for every $c \in \mathcal{C}$. If $c \leq \underline{r}$, this is immediate: $r^*(c; \kappa) \geq r^*(\underline{c}; \kappa) > \underline{r} \geq c$ by Fact 3 and the interiority condition. If $c > \underline{r}$, strict concavity of $g(\cdot, c; \kappa)$ implies that $r^*(c; \kappa) \geq c$ if and only if $g_r(c, c; \kappa) = q(c) + \kappa s^{r\prime}(c) \geq 0$. This expression is strictly decreasing in c, because $q' < 0$ on $[\underline{r}, \bar{r})$ and $s^{r\prime}$ is nonincreasing. The margin condition $r^*(\bar{c}; \kappa) \geq \bar{c}$ therefore gives $g_r(\bar{c}, \bar{c}; \kappa) \geq 0$ when $\bar{c} > \underline{r}$, and hence the same inequality at every lower cost; when $\bar{c} \leq \underline{r}$, the first case covers every c.

Lemma 1

Proof. Part (i), comparison. Fix c and write $r_i \equiv r_i^*(c)$. Revealed preference at the two optima gives

$$\begin{aligned}
(r_1 - c)q(r_1) + \kappa_1 s^r(r_1) \ &\geq\ (r_0 - c)q(r_0) + \kappa_1 s^r(r_0) \\
(r_0 - c)q(r_0) + \kappa_0 s^r(r_0) \ &\geq\ (r_1 - c)q(r_1) + \kappa_0 s^r(r_1)
\end{aligned}$$

Adding and rearranging, $(\kappa_0 - \kappa_1)\big[s^r(r_0) - s^r(r_1)\big] \geq 0$. Since $\kappa_0 > \kappa_1$, $s^r(r_0) \geq s^r(r_1)$, and since s^r is strictly decreasing, $r_1 \geq r_0$. To rule out equality, suppose $r_1 = r_0 \equiv r$. By Assumption 5 and Fact 3, $r_0^*(c) \geq r_0^*(\underline{c}) > \underline{r}$ for every c, and $r < \bar{r}$ by Fact 2, so r is interior for both configurations and satisfies both first-order conditions:

$$q(r) + (r-c)q'(r) + \kappa_1 s^{r\prime}(r) \;=\; 0 \;=\; q(r) + (r-c)q'(r) + \kappa_0 s^{r\prime}(r)$$

Subtracting, $(\kappa_0 - \kappa_1)s^{r\prime}(r) = 0$, a contradiction because $\kappa_0 > \kappa_1$ and $s^{r\prime} < 0$. Hence $r_1^*(c) > r_0^*(c)$.

Part (i), bound and limit. The monopoly rate $r^M(c) = r^*(c;0)$ is unique by strict concavity. It is interior: the revealed-preference step above, applied to the pair $(\kappa_0, 0)$, gives $r^M(c) \geq r_0^*(c) > \underline{r}$, and $r^M(c) < \bar{r}$ by Fact 2. Hence $\Pi_r(r^M(c);c) = 0$. For any $\kappa > 0$ and $r \geq r^M(c)$, $g_r(r,c;\kappa) = \Pi_r(r;c) + \kappa s^{r\prime}(r) < 0$, because $\Pi_r \leq 0$ there by strict concavity and $s^{r\prime} < 0$; hence $r^*(c;\kappa) < r^M(c)$. If $\omega^r = 0$, then $\kappa = 0$ and $r^*(c;0) = r^M(c)$. Together, $r_i^*(c) \leq r^M(c)$ with equality if and only if $\omega_i^r = 0$. For the limit, take any sequence $\kappa_n \to 0$; by continuity of r^* in κ (Fact 1), $r^*(c;\kappa_n) \to r^*(c;0) = r^M(c)$.

Part (ii). Consumer surplus: $CS_1(c) - CS_0(c) = -\int_{r_0}^{r_1} q(p)\,dp < 0$, because $[r_0, r_1] \subset [\underline{r}, \bar{r})$ by Fact 2, $q > 0$ there, and the interval has positive length by part (i). Contract value: by part (i), $r_0 < r_1 \leq r^M(c)$, and since $\Pi(\cdot;c)$ is strictly concave with maximizer $r^M(c)$, it is strictly increasing on $[\underline{r}, r^M(c)]$, so $\Pi_1(c) > \Pi_0(c)$.

Part (iii). Total surplus at rate r is $TS(r;c) = \int_r^{\bar{r}} q(p)\,dp + (r-c)q(r)$, with derivative $\partial TS/\partial r = (r-c)q'(r)$. By Fact 4, $r_0^*(c) \geq c$, and part (i) places $r_1 > r_0 \geq c$. The integrand $(p-c)q'(p)$ is nonpositive for $p \geq c$ and strictly negative for $p > c$, so

$$TS_1(c) - TS_0(c) = \int_{r_0}^{r_1} (p-c)\,q'(p)\,dp < 0,$$

the interval $(r_0, r_1]$ having positive length and lying in $(c, \bar{r})$ except possibly at the single point $r_0 = c$.

Without Assumption 5. The rate, consumer surplus, and contract value comparisons in parts (i) and (ii) hold with weak inequalities. The bound $r^*(c) \leq r^M(c)$ and the limit $r^*(c) \to r^M(c)$ as $\kappa \to 0$ remain valid, but equality with the monopoly rate can occur at positive κ when the lower bound on rates binds, so the equality characterization in part (i) fails. The total surplus comparison is indeterminate: where $r_0^*(c) < c$, raising the rate toward cost increases total surplus. $\square$

Lemma 2

Proof. Fix κ. The objective $g(r,c;\kappa)$ is affine in c with slope $-q(r) \in [-q(\underline{r}), 0]$, so $x(\cdot;\kappa)$, as a pointwise maximum of affine functions with uniformly bounded slopes, is convex and Lipschitz on $[\underline{c}, \bar{c}]$, hence absolutely continuous. Because the maximizer is unique and continuous in c (Fact 1) and the choice set does not depend on c, the envelope theorem (Milgrom and Segal, 2002) yields

$$x'(c;\kappa) = \Pi_c\big(r^*(c;\kappa);c\big) = -q\big(r^*(c;\kappa)\big)$$

on $(\underline{c},\bar{c})$, with one-sided derivatives at the endpoints, and continuity of $r^*(\cdot;\kappa)$ makes $x(\cdot;\kappa)$ continuously differentiable. By Fact 2, $q(r^*(c;\kappa)) > 0$, so $x(\cdot;\kappa)$ is strictly decreasing with $|x'(c)| = q(r^*(c))$. The argument applies at any κ, and hence to the cost component of the pseudotype in the multidimensional model. It uses neither Assumption 5 nor interiority of the rate; the identity holds when the lower bound on rates binds. $\square$

Lemma 3

Proof. Step 1: pointwise ranking of cost sensitivity. By Lemma 1(i), $r_1^*(c) > r_0^*(c)$ for every c, and both rates lie in $[\underline{r},\bar{r})$ by Fact 2, where $q' < 0$. Hence, by Lemma 2,

$$|x_1'(c)| = q(r_1^*(c)) < q(r_0^*(c)) = |x_0'(c)| \quad \text{for all } c \in \mathcal{C}.$$

Step 2: quantile representation. Both $x_0(\cdot)$ and $x_1(\cdot)$ are continuous and strictly decreasing (Lemma 2), so $X_i = x_i(C)$ has quantile function $F_{X_i}^{-1}(u) = x_i\big(F_C^{-1}(1-u)\big)$ for $u \in (0,1)$: low costs generate high pseudotypes, so the u-quantile of X_i is the image of the $(1-u)$-quantile of cost.

Step 3: dispersive order. Fix $0 < \alpha < \beta < 1$ and let $c(\beta) \equiv F_C^{-1}(1-\beta) < F_C^{-1}(1-\alpha) \equiv c(\alpha)$; the inequality is strict because F_C^{-1} is strictly increasing under Assumption 7. Then

$$F_{X_i}^{-1}(\beta) - F_{X_i}^{-1}(\alpha) = x_i(c(\beta)) - x_i(c(\alpha)) = \int_{c(\beta)}^{c(\alpha)} \big|x_i'(u)\big|\, du$$

and Step 1 gives

$$F_{X_1}^{-1}(\beta) - F_{X_1}^{-1}(\alpha) \;<\; F_{X_0}^{-1}(\beta) - F_{X_0}^{-1}(\alpha)$$

as the integrand is strictly smaller pointwise and the interval has positive length.

Step 4: order-statistic spacings. Fix $J \geq 2$ and $1 \leq m < n \leq J$. Let $U^{(1)} \geq \cdots \geq U^{(J)}$ be the order statistics of J i.i.d. uniform draws, and couple the two configurations through them: $X_i^{(j)} = F_{X_i}^{-1}(U^{(j)})$. On the event $U^{(m)} > U^{(n)}$, which has probability one, Step 3 gives

$$F_{X_1}^{-1}(U^{(m)}) - F_{X_1}^{-1}(U^{(n)}) \;<\; F_{X_0}^{-1}(U^{(m)}) - F_{X_0}^{-1}(U^{(n)})$$

Taking expectations orders every expected spacing strictly; the case $(m,n) = (1,2)$ is the expected gap between the best and second-best pseudotypes.

Without Assumption 5. The interiority condition enters only through the strictness of Lemma 1(i); without it, rates, quantile spreads, and spacings are ordered weakly. $\square$

Theorem 1

Proof. By Lemma 2, each $x_i(\cdot)$ is strictly decreasing, so in both configurations the auction is won by the lowest-cost bidder, whose cost $c^w \equiv \min_j c_j$ has a distribution that does not depend on the configuration.

Part (i). Conditional on $c^w = c$, Lemma 1(i) gives $r_1^*(c) > r_0^*(c)$, and Lemma 1(ii)â(iii) give $CS_1(c) < CS_0(c)$ and $TS_1(c) < TS_0(c)$. Taking expectations over the fixed distribution of c^w preserves the strict inequalities.

Part (ii). By the equilibrium characterization and revenue equivalence (Appendix A.1), expected winner profit equals the expected gap between the best and second-best pseudotypes, $E[\pi_i] = E\big[X_i^{(1)} - X_i^{(2)}\big]$, which is strictly smaller at $i = 1$ by Lemma 3. Expected contract value $E[\Pi_i(c^w)]$ is strictly larger at $i = 1$ by Lemma 1(ii) and the fixed winner distribution. The winner's commission is $k = \Pi - \pi$, so

$$E[k_1] - E[k_0] = \big(E[\Pi_1] - E[\Pi_0]\big) + \big(E[\pi_0] - E[\pi_1]\big) > 0$$

with both terms strictly positive: the gain in expected commission equals the gain in expected contract value plus the loss in expected profit. □

Proposition 1

Proof. Part (i). By (A.4), the equilibrium rate maximizes $\Pi(r; c) + \psi(r)$, an objective in which neither J nor rivals' types appear. Hence $r^*(c)$ does not depend on J.

Part (ii). By (A.12), the equilibrium normalized score bid of a firm with pseudotype $x > \underline{t}$ facing $J - 1$ rivals is

$$b_J(x) = x - \int_{\underline{t}}^{x} \left[\frac{T(u)}{T(x)}\right]^{J-1} du,$$

with $b_J(\underline{t}) = \underline{t}$ for every J, and the pseudotype distribution T does not depend on J. The distribution T is atomless (Appendix A.1), hence continuous with $T(\underline{t}) = 0$, and $T(x) > 0$ for every $x > \underline{t}$ because $\underline{t}$ is the infimum of the support. Fix $x > \underline{t}$. Since T is nondecreasing, the ratio $T(u)/T(x)$ lies in $[0, 1]$ for all $u \in (\underline{t}, x)$, so the integrand is nonincreasing in J pointwise; and by continuity of T at $\underline{t}$, there is a nondegenerate interval $(\underline{t}, u_0)$ on which $0 < T(u) < T(x)$, where the integrand is strictly decreasing in J. The shading integral therefore strictly decreases in J, so $b_J(x)$ is strictly increasing in J, and so is the commission bid $k_J^* = b_J(x) - \psi(r^*, v)$ by (A.2), because $\psi(r^*, v)$ does not depend on J: the rate by part (i), and the quality score as a component of the firm's type. At $x = \underline{t}$, the bid equals $\underline{t}$ for every J, so the comparison is weak at the lower endpoint. Because T is atomless, the pseudotype exceeds $\underline{t}$ with probability one, and commission bids are strictly increasing in J almost surely. The argument does not require T to be strictly increasing, and so accommodates atoms or gaps in F_V, as permitted by Assumption 7.

Part (iii). In the model of Section 3.2 the winner is the lowest-cost bidder (Lemma 2), whose cost is $C_{(1:J)} = \min_j c_j$. For $J' > J$, $\Pr(C_{(1:J)} > c) = [1 - F_C(c)]^J$ is strictly decreasing in J on the interior of the support, so $C_{(1:J')}$ is strictly smaller in the usual stochastic order. Under Assumption 5 and Fact 3, rates are interior for every cost, so $r^*(\cdot)$ is strictly increasing (Fact 3), and consumer surplus is strictly decreasing in the rate, so $c \mapsto -CS(r^*(c))$ is strictly increasing. Total surplus satisfies $\partial TS/\partial c = -q(r^*(c)) < 0$ and $\partial TS/\partial r = (r - c)\, q'(r) \leq 0$ at $r = r^*(c) \geq c$ (Fact 4), so $c \mapsto -TS(r^*(c); c)$ is strictly increasing. First-order stochastic dominance then gives $E[r^*(C_{(1:J')})] < E[r^*(C_{(1:J)})]$ and the reverse strict inequalities for expected consumer surplus and expected total surplus. □

Proposition 2

Proof. Let the weights vary along a differentiable path with $d\omega^k > 0$, $d\omega^r \leq 0$, and $d\omega^v \leq 0$; because ω^k is strictly increasing along the path, we parameterize by ω^k. By the envelope theorem applied to (4), since r^* maximizes $\Pi(r; c) + \psi(r, v)$, the sensitivity of the pseudotype to cost is $\partial x/\partial c = \Pi_c(r^*; c)$. Under Assumption 4, $\Pi_c(r; c) = -q(r)$, so $|\partial x/\partial c| = q(r^*(c))$. From (8), the sensitivity to the quality score is $\partial x/\partial v = \omega^v/(\omega^k \alpha)$. The ratio of cost to quality score sensitivity is therefore

$$R(c; \omega^k) \equiv \frac{|\partial x/\partial c|}{\partial x/\partial v} = \frac{q(r^*(c))\, \omega^k\, \alpha}{\omega^v}.$$

The derivative $dr^*/d\omega^k$ exists by the implicit function theorem applied to the rate first-order condition, because rates are interior under Assumption 5, the rate objective is strictly concave, and s^r is twice continuously differentiable under Assumption 6; it equals $-\big(s^{r\prime}(r^*)/g_{rr}\big)\, d\kappa/d\omega^k \geq 0$, where $g_{rr} = \Pi_{rr} + \kappa s^{r\prime\prime} < 0$, because $s^{r\prime} < 0$ and $\kappa = \omega^r/(\omega^k \alpha)$ is nonincreasing along the path. Differentiating R along the path:

$$\frac{dR}{d\omega^k} = \frac{\alpha}{\omega^v}\left[q(r^*) + \omega^k\, q'(r^*)\frac{dr^*}{d\omega^k}\right] - \frac{q(r^*)\, \omega^k\, \alpha}{\omega^{v\,2}}\frac{d\omega^v}{d\omega^k}$$

Dividing through by $q(r^*)\, \alpha/\omega^v > 0$, $dR/d\omega^k > 0$ if and only if

$$1 - \frac{|q'(r^*)|\, \omega^k}{q(r^*)}\frac{dr^*}{d\omega^k} + \frac{\omega^k}{\omega^v}\left|\frac{d\omega^v}{d\omega^k}\right| > 0,$$

using $q' < 0$ and $dr^*/d\omega^k \geq 0$ to write the middle term as a magnitude, and $d\omega^v/d\omega^k \leq 0$ for the last. Because $dr^*/d\omega^k \geq 0$, the middle term equals $\varepsilon_q(c)\, \varepsilon_r(c)$ wherever $r^*(c) \neq 0$, upon multiplying and dividing by r^*, and the last term is ε_v, so the condition is

$$\varepsilon_q(c)\, \varepsilon_r(c) < 1 + \varepsilon_v$$

as stated. □

A.3 Equilibrium with Predetermined Commissions

Section 3.4 states the bidder's problem when the commission is predetermined ($\omega_i^k = 0$). Here we describe the iterative best-response algorithm we use to approximate equilibrium (e.g., Bajari, 2001; Armantier et al., 2008). The algorithm parameterizes a "belief function" that determines how firms expect their competitors to bid, computes the best responses to those beliefs, updates the belief function based on the best responses, and iterates until the beliefs are consistent with the fitted best responses. The converged belief function is the approximate equilibrium strategy. In ICS, we find a tight relationship between the belief functions and the best-response functions, indicating that the approximation is accurate in practice.

To implement the approximation, we assume that in each iteration, $h = 0, 1, \ldots, H$, all firms believe rival j in auction i will bid according to

$$\hat{s}_{ij}^{(h)} = g_i\left(\mathbf{x}_{ij}; \gamma_i^{(h)}\right) \tag{A.14}$$

where $g_i()$ is the belief function, the auction-specific vector $\gamma_i^{(h)}$ contains "belief coefficients," and the vector $\mathbf{x}_{ij}$ includes exogenous factors. We use the rival's cost and technical score, plus a constant; since $g_i()$ is auction-specific, the constant absorbs factors that do not vary across bidders (e.g., the auction weights). Given these beliefs, we compute the best responses, $\hat{s}_{if}^{(h)BR}$, of $f \in \{1, \ldots, F\}$ simulated "focal" firms by solving their profit maximization problem. For each auction, we regress these focal firms' best responses on a function of their exogenous factors to obtain the regression coefficients, $\hat{\gamma}_i^{(h)}$. We then update the belief coefficients using

$$\gamma_i^{(h+1)} = \rho^{(h)}\hat{\gamma}_i^{(h)} + \left(1 - \rho^{(h)}\right)\gamma_i^{(h)} \tag{A.15}$$

where $\rho^{(h)} \in [0, 1]$ controls the step size; we reduce the step size with the number of iterations to promote convergence. We run 21 iterations using linear belief functions and then 21 iterations in which $g_i()$ is a second-order polynomial in the exogenous factors, warm-started from the converged linear coefficients, with zeros appended for the higher-order terms.

Evaluated at our final estimates of the structural parameters, the average mean squared error between the belief coefficients and the regression coefficients is only 0.0008 after the 42 iterations. Across auctions, the average R^2 in the final regression is 0.9566. Thus, the belief coefficients are close to a fixed point of the updating map, and the polynomial specification captures most of the variation in best responses across firm types. We provide details and results in Appendix B.3.

B Empirical Model and Estimation

B.1 Identification

We establish identification of the model primitives from the procurement record. The econometrician observes, for each auction i with $J_i \geq 2$ bidders, the scoring weights $(\omega_i^r, \omega_i^k, \omega_i^v)$, the rate and commission bids (r_{ij}, k_{ij}) of every bidder, and the technical score assigned to every bidder. The demand function $q(\cdot)$ and the contribution functions are identified in prior steps and treated as known here. Identification is nonparametric for the cost distribution F_C. For technical scores, we maintain the location-scale structure $v_{ij} = \mu_i + \sigma\,\eta_{ij}$, where η_{ij} is i.i.d. with mean zero and unit variance. The auction-level locations μ_i play no role in equilibrium bids: rates do not depend on technical scores, commissions depend on them only through the information rent, which is invariant to a common location shift of the pseudotypes, and a common shift in v shifts all scores equally, leaving rankings unchanged. We therefore normalize $\mu_i = 0$ without loss of generality and work with within-auction demeaned technical scores throughout. For F_η we use the standardized empirical distribution of demeaned technical scores directly, so that no parametric family is imposed on the shape. Although this distribution is discrete, the pseudotype distribution is absolutely continuous regardless, because its cost component is: a sum that includes an independent absolutely continuous term has a density, which is all that the argument below uses.[30] The within-auction scale σ is identified from equilibrium rate and commission moments, without reference to the dispersion of the technical score data. The score data thus supply the shape of the technical score distribution, whereas equilibrium bids supply its scale. Remark 1 discusses the overidentification this implies and our treatment of it; the location-scale structure itself, with a common σ and common standardized shape across auctions, is a pooling restriction motivated by the size of the bid sample.

Let $f_i(c) \equiv \frac{\omega_i^r}{\omega_i^k}\frac{1}{\alpha}\, s_i^r(r^*(c)) + \Pi(r^*(c); c)$ denote the cost component of the pseudotype, so that $x_i^*(c, v) = f_i(c) + \frac{\omega_i^v}{\omega_i^k \alpha}\, v$ by equation (8), and write $a_i \equiv \omega_i^v/(\omega_i^k \alpha)$.

Proposition B.1 (Identification)**.** *Maintain Assumptions 3, 4, 6, and 7, together with the interiority condition of Assumption 5 ($r_i^*(\underline{c}) > \underline{r}$ and $r_i^*(\bar{c}) \geq \bar{c}$) in the auctions used for the inversion, with $v_{ij} = \sigma\,\eta_{ij}$ after the location normalization, $\sigma > 0$, and F_η given. Then: (i) F_C is identified from observed rates in auctions with $\omega_i^r > 0$, $\omega_i^k > 0$, and rates not predetermined by the buyer; and (ii) if some auction i satisfies the conditions of part (i) and, in addition, $\omega_i^v > 0$, then σ is identified from the population mean of $\Pi(r_{ij}^*; c_{ij}) - k_{ij}^*$ across the bids of that auction.*

The remainder of this subsection proves the proposition in two steps.

[30] The location-scale structure with auction-specific locations modifies Assumption 7, which specifies a common distribution F_V; the assumption's remaining requirements—bounded support and independence of cost and technical score—are maintained throughout, and absolute continuity holds for the pseudotype distribution, which is what the argument uses.

Step 1: Cost Distribution, $F_C(\cdot)$

As established in Appendix A.1, observed equilibrium rates are set to maximize $\Pi(r;c) + \psi_i(r,v)$. Given the contract value form, $\Pi_r(r_{ij};c_{ij}) = q(r_{ij}) + (r_{ij} - c_{ij})q\prime(r_{ij})$, so that the rate first-order condition (A.6) obtains, with $\kappa_i \equiv \omega_i^r/(\omega_i^k \alpha)$ as defined in (A.4):

$$c_{ij} = r_{ij} + \big(q(r_{ij}) + \kappa_i\, s_i^r\prime(r_{ij})\big)\big(q\prime(r_{ij})\big)^{-1} \tag{B.1}$$

All objects on the right-hand side of this equation are observed or estimable in the data. Given an interior solution, the first-order condition is necessary and, by strict concavity of the rate objective, sufficient; because $q\prime(r) < 0$, it defines a unique mapping from the observed rate r_{ij} to c_{ij}. Hence, bidder-level costs can be recovered from observed rates, and pooling these across auctions with $\omega_i^k > 0$, $\omega_i^r > 0$, and rates not predetermined by the buyer identifies the unconditional cost distribution $F_C(\cdot)$ nonparametrically. The restriction to $\omega_i^k > 0$ is required because the first-order condition (A.6) rests on the triangular structure established in Appendix A.1: with the commission available to absorb score differences, the rate bid is non-strategic and the closed-form inversion applies. When $\omega_i^k = 0$, or when the buyer predetermines the commission, the rate bid depends on the distribution of competing bids and costs cannot be recovered by this inversion; these auctions are excluded from the cost inversion but remain informative at the distribution level (Remark 3).

Step 2: Scale of Technical Scores, σ

Combining equations (A.12) and (A.13), the equilibrium commission satisfies

$$k_{ij}^* = \Pi(r_{ij}^*;c_{ij}) - \mathrm{IR}_i(x_{ij}^*;\sigma), \qquad \mathrm{IR}_i(x;\sigma) \equiv \frac{\int_{\underline{t}}^{x} T_i(u;\sigma)^{J_i-1}\,du}{T_i(x;\sigma)^{J_i-1}}, \tag{B.2}$$

because $x_{ij}^* - \psi_i(r_{ij}^*, v_{ij}) = \Pi(r_{ij}^*;c_{ij})$ by (8). The technical score enters the score bid only through the pseudotype and enters ψ_i directly with the same multiplier a_i, so the direct contributions cancel and the commission depends on the technical score only through the information rent. Accordingly, $\Pi(r_{ij}^*;c_{ij}) - k_{ij}^* = \mathrm{IR}_i(x_{ij}^*;\sigma)$ is recovered for every bid from observed commissions and the costs identified in Step 1. Conditional on cost, the commission is decreasing in the technical score whenever the information rent is increasing in the pseudotype, as higher-v firms then earn larger rents.

The pseudotype distribution, replacing $v = \sigma\eta$ in equation (A.11), is

$$T_i(x;\sigma) = \iint \mathbf{1}(x_i^*(c,\,\sigma\eta) \le x)\, dF_C(c)\, dF_\eta(\eta), \tag{B.3}$$

where the product form of the integrator uses the independence of c and v maintained in Assumption 7. Independence is important for three points in this step: the product integral in (B.3), the variance decomposition in (B.4) below, and the convolution argument in the proof of Lemma B.1.

Given F_C from Step 1, F_η, and the known weights and contribution functions, $T_i(\cdot\,;\sigma)$ and hence $\mathrm{IR}_i(\cdot\,;\sigma)$ are known functionals of σ alone. Since $x_i^* = f_i(C) + a_i\sigma\eta$ with $C \perp \eta$ and $\mathrm{Var}(\eta) = 1$, the pseudotype variance is

$$\mathrm{Var}(x_i^*; \sigma) = \mathrm{Var}(f_i(C)) + a_i^2\,\sigma^2, \tag{B.4}$$

which is strictly increasing in σ whenever $\omega_i^v > 0$; this strictness is used in the proof of Lemma B.1.

Under the model taken at face value, the pseudotypes are computable bid by bid: with costs recovered in Step 1 and the constructed score measure equated with v_{ij}, $x_{ij}^* = f_i(c_{ij}) + \frac{\omega_i^v}{\omega_i^k \alpha}\, v_{ij}$ requires no knowledge of σ, and one could trace IR_i as an observed function of its argument by pairing each recovered rent with its computed pseudotype. We deliberately do not base identification of σ on this pairing, for two reasons. If the bidder-level score measure is taken at face value, σ is identified directly by the within-auction dispersion of the scores, and a bid-based argument is redundant. If instead the score measure is a distorted observation of v_{ij}—the interpretation adopted in Remark 1—then the computed pseudotypes inherit the distortion and the pairing is contaminated. In either case, an identification argument for σ from bids that is robust to this measurement concern must operate on moments of the recovered rent values $\Pi(r_{ij}^*; c_{ij}) - k_{ij}^*$, which are computed from rates and commissions alone; it touches the score data only through the aggregate shape of F_η. Identification of σ from bid moments then requires that distinct values of σ generate distinct observable distributions of these rent values. The following lemma establishes more: the population mean of the recovered rents is strictly increasing in σ, so a single first moment of observed commissions identifies the scalar parameter, and the remaining bid-level and cross-auction moments overidentify it.

A preliminary observation addresses the rent transformation itself. Were pseudotypes paired with rents in the data, the rent function would identify the pseudotype distribution directly: writing $G_i(x) \equiv \int_{\underline{t}}^{x} T_i(u;\sigma)^{J_i-1} du$, equation (B.2) gives $(\log G_i)'(x) = 1/\mathrm{IR}_i(x;\sigma)$, a first-order ordinary differential equation that determines G_i up to a multiplicative constant, which is pinned by $T_i(\bar{t};\sigma) = 1$; two pseudotype distributions on a common support with the same rent function therefore coincide. The transformation from distributions to rent functions thus loses no information. Because pseudotypes are not paired with rents in our data, however, identification cannot exploit this inversion pointwise and instead runs through the moment established next.

Lemma B.1. *In auction i, suppose the pseudotype satisfies $x_i^* = W_i + a_i\sigma\eta$, where $a_i > 0$, η has mean zero, unit variance, and bounded support, W_i is independent of η with bounded support, and, for every σ under consideration, $x_i^*(\sigma)$ has an absolutely continuous distribution with a strictly positive density on a compact interval. Then $\mathbb{E}[\mathrm{IR}_i(x_i^*;\sigma)]$ is strictly increasing in σ. In particular, the conclusion holds with $W_i = f_i(C)$ when $\omega_i^k > 0$ and $\omega_i^v > 0$; the regularity condition holds in our application, since $f_i(C)$ is a strictly monotone transform of a cost with a strictly positive density.*

Proof. Let $x_\sigma(\cdot)$ denote the quantile function of $T_i(\cdot\,;\sigma)$, absolutely continuous and strictly increasing on its support by the regularity hypothesis. The change of variables $u = x_\sigma(q)$ in (B.2) gives

$\mathrm{IR}_i(x_\sigma(p);\sigma) = p^{-(J_i-1)}\int_0^p q^{J_i-1}x'_\sigma(q)\,dq$, and since the quantile rank of each bidder's pseudotype is uniform on $(0,1)$,

$$\mathbb{E}[\mathrm{IR}_i(x^*;\sigma)] = \int_0^1 x'_\sigma(u)\,w_i(u)\,du, \qquad w_i(u) \equiv u^{J_i-1}\int_u^1 p^{-(J_i-1)}dp.$$

Since $w_i(0) = w_i(1) = 0$ and x_σ is bounded, integration by parts yields the L-functional representation

$$\mathbb{E}[\mathrm{IR}_i(x^*;\sigma)] = \int_0^1 x_\sigma(u)\,\varphi_{J_i}(u)\,du, \qquad \varphi_{J_i} \equiv -w'_i, \tag{B.5}$$

where φ_{J_i} is strictly increasing with $\int_0^1 \varphi_{J_i} = 0$: for $J_i \geq 3$, $w_i(u) = (u - u^{J_i-1})/(J_i-2)$ and $\varphi_{J_i}(u) = \big((J_i-1)u^{J_i-2}-1\big)/(J_i-2)$; for $J_i = 2$, $w_i(u) = -u\log u$ and $\varphi_2(u) = 1 + \log u$.

An L-functional with an increasing, mean-zero weight is monotone in the convex order among distributions with equal means. Using $\int_0^1 \varphi_{J_i} = 0$ and the representation $\varphi_{J_i}(u) = \int_{(0,1)} \big[\mathbf{1}\{u \geq s\} - (1-s)\big]\,d\varphi_{J_i}(s)$ for almost every u, Fubini's theorem gives

$$\int_0^1 x_\sigma(u)\,\varphi_{J_i}(u)\,du = \int_{(0,1)} \Big[\int_s^1 \big(x_\sigma(u) - m_i\big)\,du\Big]\,d\varphi_{J_i}(s), \tag{B.6}$$

where $m_i \equiv \mathbb{E}[x_i^*]$ does not vary with σ because $\mathbb{E}[\eta] = 0$. The inner integral is $O(s)$ as $s \to 0$ by boundedness of the support, so (B.6) is finite also for $J_i = 2$, where $d\varphi_2(s) = ds/s$.

It remains to order the upper quantile integrals in σ. For $\sigma < \sigma'$, write $a_i\sigma\eta = \theta\,(a_i\sigma'\eta) + (1-\theta)\cdot 0$ with $\theta = \sigma/\sigma' \in (0,1)$. For any convex ψ, convexity and then Jensen's inequality (using $\mathbb{E}[\eta] = 0$) give $\mathbb{E}[\psi(a_i\sigma\eta)] \leq \theta\,\mathbb{E}[\psi(a_i\sigma'\eta)] + (1-\theta)\,\psi(0) \leq \mathbb{E}[\psi(a_i\sigma'\eta)]$, so $a_i\sigma\eta \leq_{\mathrm{cx}} a_i\sigma'\eta$. The convex order is closed under convolution with an independent random variable (Shaked and Shanthikumar, 2007, Theorem 3.A.12), so $x_i^*(\sigma) \leq_{\mathrm{cx}} x_i^*(\sigma')$ with equal means; the independence of W_i and η is essential at this step. Equal-mean convex ordering is equivalent to $\int_s^1 x_\sigma(u)\,du \leq \int_s^1 x_{\sigma'}(u)\,du$ for all $s \in (0,1)$ (Shaked and Shanthikumar, 2007, Section 3.A), so (B.6) is weakly increasing in σ.

Strictness follows from the variance. $\mathrm{Var}(x_i^*;\sigma) < \mathrm{Var}(x_i^*;\sigma')$ by (B.4), so the two distributions differ; were the upper quantile integrals equal for almost every s, the quantile functions would coincide almost everywhere and the variances would be equal. The set of s on which the inequality is strict therefore has positive measure, and since $d\varphi_{J_i}$ has a strictly positive density on $(0,1)$—$(J_i-1)u^{J_i-3}$ for $J_i \geq 3$ and $1/u$ for $J_i = 2$—the right-hand side of (B.6) is strictly larger under σ' than under σ. □

Lemma B.1 completes the proof of Proposition B.1(ii): the population mean of $\Pi(r_{ij}^*; c_{ij}) - k_{ij}^*$ across bids in auction i equals $\mathbb{E}[\mathrm{IR}_i(x^*;\sigma)]$, a known and strictly increasing function of σ, so this moment identifies σ. The proposition concerns the population moment; in estimation, the corresponding sample moment pools bids within and, under the maintained common σ, across auctions, so precision is not limited by the number of bids in any single auction. Representation

(B.5) also makes the moment interpretable: for $J_i \geq 3$ it reduces to the closed form

$$\mathbb{E}\big[\Pi(r_{ij}^*; c_{ij}) - k_{ij}^*\big] = \frac{\mathbb{E}\big[x^{(1:J_i-1)}\big] - \mathbb{E}[x_i^*]}{J_i - 2},$$

where $x^{(1:J_i-1)}$ denotes the maximum of $J_i - 1$ independent draws from $T_i(\cdot\,; \sigma)$ (for $J_i = 2$, $\mathbb{E}[\mathrm{IR}_i] = \mathbb{E}[x_i^*] + \mathbb{E}[x_i^* \log T_i(x_i^*; \sigma)]$, with the same structure). Greater within-auction dispersion in technical scores spreads the pseudotype distribution and raises the expected excess of the best of $J_i - 1$ rival pseudotypes over the mean pseudotype—the statistic that governs equilibrium rents—so observed commissions, which equal contract value net of rents, are correspondingly lower.[31]

Although the population commission moment in a single auction suffices for identification, cross-auction variation supplies overidentifying content that the estimator exploits and that underlies the use of the scoring-rule weights as instruments in equations (16)–(17). Variation in J_i changes the known L-functional that maps the pseudotype distribution into mean bid shading—for $J_i \geq 3$, $\mathbb{E}[\mathrm{IR}_i] = \big(\mathbb{E}[x_i^{(1:J_i-1)}] - \mathbb{E}[x_i]\big)/(J_i - 2)$—so auctions with different numbers of bidders provide different mappings from the common σ to observed commissions; no monotonicity of the first–second spacing in J_i is needed. Variation in $a_i = \omega_i^v/(\omega_i^k \alpha)$ shifts the slope of $\mathbb{E}[\mathrm{IR}_i(\cdot\,; \sigma)]$ with respect to σ across auctions—see equation (B.4)—which is why the weights serve as instruments: they generate predictable cross-auction variation in the mapping from σ to observed commissions. The validity of this variation requires that the scoring weights be orthogonal to the primitives (F_C, F_η, σ), the same restriction implicit in pooling F_C across auctions. In addition, the rank-specific rate moments used in estimation vary with σ, because the ranking of bidders by score depends on the relative dispersion of costs and technical scores, and the rate moments of auctions with predetermined commissions depend on σ directly, because rate bids there are strategic (Remark 3); these moments provide further identifying content and contribute to the precision of $\hat{\sigma}$.

Remark 1 (Overidentification and the technical score data). The technical score data identify σ directly: under the model, the within-auction standard deviation of demeaned technical scores converges to σ. Equilibrium bids identify σ independently, through Lemma B.1. Our estimator resolves this overidentification in favor of the bid moments, so the estimate should be interpreted as the dispersion of the technical score as it operates in equilibrium bidding. Any excess dispersion in the constructed measure can be attributed to its construction: the measure aggregates scoring elements related to technical capabilities and subjective assessments and normalizes by maximum possible points across heterogeneous categories, which can inflate cross-bidder dispersion relative to the score object in the model. The same measurement concern dictates the design of Step 2: although the pseudotypes would be computable bid by bid if the score measure were equated

[31] The selection logic is transparent in the closed form: $\partial\, \mathbb{E}[x^{(1:J_i-1)}]/\partial\sigma = a_i\, \mathbb{E}[\eta_{\mathrm{argmax}}] > 0$, because the maximizing draw is positively selected on η, while $\mathbb{E}[x_i^*]$ does not vary with σ. Note that the relevant statistic is the best of $J_i - 1$ draws relative to the mean, not the gap between the first- and second-best pseudotypes, which is a different moment.

with v_{ij}, our identification of σ never pairs recovered rents with pseudotypes computed from the bidder-level score measure, and so remains valid when that measure is distorted.

Remark 2 (Auctions with predetermined rates). When the buyer predetermines the rate at $\bar{r}_i$ and $\omega_i^k > 0$, Step 1 is unavailable and bidder-level costs are not recovered. These auctions nonetheless contribute distribution-level information. The rate-score term is common across bidders, so the pseudotype reduces to $x_i^* = \Pi(\bar{r}_i; C) + a_i\sigma\eta$ up to a common constant, and equation (B.2) holds with $\bar{r}_i$ in place of r_{ij}^*. With F_C identified from the remaining auctions, $\mathbb{E}[k_{ij}^*] = \mathbb{E}[\Pi(\bar{r}_i; C)] - \mathbb{E}[\mathrm{IR}_i(x^*; \sigma)]$ is known up to σ and, by Lemma B.1 applied with $W_i = \Pi(\bar{r}_i; C)$—under the regularity condition of Lemma B.1—strictly decreasing in σ. Expected commissions in these auctions therefore overidentify σ.

Remark 3 (Auctions with predetermined commissions). The mirror case is a buyer who predetermines the commission while the rate is bid. With the commission unavailable to absorb score differences, the triangular structure fails and the rate bid is strategic: the optimal rate trades off contract value against the probability of winning. Bidder-level costs are therefore not recovered by the inversion of Step 1, and the predetermined commissions do not provide an endogenous commission-bid moment. These auctions nonetheless contribute distribution-level information through their rates: the equilibrium rate function depends on the distribution of rivals' equilibrium score bids, which is induced by (F_C, F_η, σ), so with these identified from the auctions covered by Proposition B.1, observed rates in these auctions supply overidentifying moments—note that, unlike the non-strategic rates of Step 1, these rates carry information about σ as well as costs. In estimation they accordingly enter only the rate moments, with model rates computed from the numerical equilibrium approximation of Appendix B.3.

Remark 4 (Role of the primitives). Identification of (F_C, σ), rather than of the pseudotype distributions T_i alone, is what permits counterfactual analysis. A change in the scoring weights alters the mapping $x_i^*(c, v)$ and hence T_i; reconstructing equilibrium under counterfactual weights requires the underlying cost and technical score distributions, which is why the pseudotype distribution implied by observed bids does not suffice.

B.2 Integrated Splines and the Cost Distribution

An integrated spline (I-spline) is a linear combination of basis functions. Each basis function is continuously differentiable and non-decreasing, mapping $[0, 1]$ to $[0, 1]$. Recycling Q and p from a notational perspective, the I-spline takes the form

$$Q(p) = \tau_0 + \sum_{j=1}^{K} \tau_j I_{j,d}(p) \tag{B.7}$$

where $\tau_0, \ldots, \tau_K$ are the parameters we estimate and $I_{j,d}(p)$ are the basis functions. Thus, the I-spline is continuously differentiable and maps $[0, 1]$ to an unrestricted range on the real line. We restrict the weights $(\tau_1, \ldots, \tau_K)$ to be nonnegative, which ensures that the I-spline is nondecreasing and therefore a *quantile function*, or inverse CDF; strictly positive weights make it strictly increasing, so that the implied distribution has the strictly positive density required by Assumption 7, as holds at our estimates. In our application, p is a probability level and $Q(p)$ is the corresponding cost quantile. Figure D.7 shows the quantile function and the implied CDF that we estimate. Drawing $p \sim \text{Uniform}(0, 1)$ and computing $c = Q(p)$ generates a random cost draw from $F_C(c) = Q^{-1}(c)$.

The flexibility of the I-spline depends on the knot placement (which controls where curvature changes) and the polynomial degree d (which controls smoothness). We use $d = 1$ with one interior knot at $p = 0.50$, which yields $K = 3$ basis functions and four parameters to estimate: $\{\tau_0, \tau_1, \tau_2, \tau_3\}$. Our specification allows the quantile function to be globally convex, globally concave, locally concave then locally convex, or locally convex then locally concave; the CDF has the opposite curvature at each point, because $F_C''(c) = -Q''(p)/Q'(p)^3$ at $p = F_C(c)$. Figure D.8 illustrates this flexibility using different parameterizations. The I-spline can be made more or less flexible; more flexibility requires more parameters.

The I-spline basis functions $I_{j,d}(p)$ are constructed through a three-step process. First, B-splines of degree d are defined recursively over a knot sequence $\{\xi_1, \ldots, \xi_{K+d+1}\}$. These are piecewise polynomials, each with compact support. Second, M-splines are obtained by normalizing B-splines so they integrate to one over their support. Because B-splines are nonnegative, M-splines are also nonnegative. Third, I-spline basis functions are obtained by integrating M-splines:

$$I_{j,d}(p) = \int_{\xi_1}^{p} M_{j,d}(t)\, dt. \tag{B.8}$$

Since M-splines are nonnegative, each I-spline basis function is nondecreasing. Integration increases the polynomial degree by one, so with $d = 1$ (linear M-splines), the I-spline basis functions are piecewise quadratic. For more details, see de Boor (2001) and Schumaker (1981).

B.3 Equilibrium Approximation

For each candidate parameter vector evaluated by the optimizer, we approximate equilibrium in the auctions for which the commission does not enter the scoring rule, as described in Appendix A.3. We use 21 iterations with a linear specification of the belief function and use the results to warm-start 21 iterations with a second-order polynomial specification. Here, we provide implementation details and then summarize the results, first for the intermediate linear beliefs and then for the final polynomial beliefs.

Details on Implementation

We take ten evenly-spaced draws from both the cost distribution and the within-auction technical score distribution.[32] The combination of these draws provides 100 representative "focal firms" for which we obtain best responses given a set of beliefs. We compute these focal firms' best responses numerically. For a candidate score bid from a focal firm with a known cost and technical score, we invert the scoring rule to obtain the implied rate and compute profit conditional on winning, net of the predetermined commission. We obtain the probability of winning as the fraction of 1,000 simulated competitor sets whose bids, given the beliefs, all fall below the candidate score. Competitors are drawn from a panel of 1,001 types that pairs evenly-spaced cost percentiles with independent draws from the standardized distribution of technical score residuals; each set draws $J_i - 1$ types without replacement. Thus, we consider a full range of possible competitors; the focal firms are not restricted to competing only with each other. The candidate bid that maximizes expected profit is the best response.[33] We update beliefs as described above, with $\rho^{(h)} = 0.50$ in iterations two through seven, 0.10 in iterations one and eight through fourteen, and 0.01 thereafter, in both the linear and the polynomial stage. The process converges reliably in practice.

Detailed Results

Table D.3 shows the regression results obtained in the final iteration with linear beliefs. The dependent variable is the focal firms' best response; the regressors are their cost and technical score. Each row shows the results for a single auction. In each auction, the focal firms' score bids decrease with cost and increase with technical score, consistent with profit-maximizing behavior. The results show that the best responses are well approximated with a linear function: the average R^2 value is 0.9198. The polynomial specification improves on this, as we discuss below. The mean squared error between belief and regression coefficients, averaged across auctions, is 0.0007. This is small relative to the magnitude of the coefficients, indicating alignment between the best responses and the beliefs.

The heterogeneity in belief coefficients across auctions reflects differences in auction design and competitive intensity. To illustrate, Table D.4 regresses each belief coefficient on the auction weights and the number of bidders. The cost coefficient γ_1 becomes more negative when the rate weight is larger and when there are more competitors, meaning that best responses are more sensitive to costs in auctions where rates matter more and competition is fiercer. The technical score coefficient γ_2 increases with the technical weight and the number of bidders, as expected. The constant γ_0 is not strongly explained by these covariates ($R^2 = 0.18$), consistent with auction-specific factors such as the scoring rule intercepts absorbing much of the level variation. These patterns align with

[32] That is, we take the 5th percentile, 15th percentile, 25th percentile, and so on, up to the 95th percentile. Thus, the draws are evenly distributed in probability space.

[33] We use the optimize function in R, which requires lower and upper bounds. For each focal firm, we use the scores implied by its monopoly rate and its at-cost rate.

the model's comparative statics and are captured by our auction-specific belief functions.

Table D.5 shows the belief coefficients, R^2 values, and MSE values after all 42 iterations. The polynomial specification improves the fit: the average R^2 increases from 0.9198 to 0.9566. The average MSE between belief and regression coefficients is 0.0008. As these are the final beliefs, we treat them as the approximate equilibrium strategies and use them to generate the simulated bids that enter the moment conditions.

B.4 Robustness of Estimation Results

We examine the demand specification and the linearity of the contribution functions.

Demand for Calls

We explore the sensitivity of the cost distribution to demand estimation by shifting the demand intercept upward to reflect a \$1.00 increase in willingness to pay for calls. Econometrically, the implications of higher demand for cost distribution estimates are ambiguous. On one hand, higher demand requires higher costs to rationalize observed commissions. On the other hand, lower costs are required to rationalize observed rates. Figure D.5 plots the cost distribution from the baseline model and the alternative with higher demand. We find that the cost distribution shifts toward higher costs with the alternative demand function. Thus, the commission channel dominates empirically. The mean cost under the alternative CDF is \$0.67, and the average cost of the winning bidder is \$0.39. The model fit deteriorates under the alternative demand function—the objective function increases from 0.058 to 0.155.

Contribution Functions

We examine the linearity of the contribution functions by plotting the residuals from columns (ii), (iii), and (iv) of Table 3 against the proposed contract term, together with quadratic fits, in Figure D.6. The relationship between the residuals and the quadratic term is weak. If the quadratic term is added directly to the contribution functions, the R^2 values increase only trivially: from 0.910 to 0.919, 0.984 to 0.985, and 0.972 to 0.974. Therefore, we view linearity as being a reasonable restriction in our setting. Although our empirical methods extend to nonlinear contribution functions, optimal rate bids would no longer have a closed-form expression.

C Empirical Evidence on the Effect of Rates on Call Activity

We exploit two natural experiments to estimate call demand. Both are based on state-level policy changes. In 2010, New York eliminated a \$1.28 connection charge and reduced the per-minute price from \$0.068 to \$0.048. The implied rate of a 15-minute call fell from \$2.30 to \$0.72. New

Jersey implemented a series of price reductions from January 2014 to May 2015 that lowered the per minute price from \$0.33 to \$0.044 (from \$4.95 to \$0.66 for a 15-minute call).[34]

To estimate demand, we regress the number of calls on the price of a 15-minute call, using data from New Jersey and New York before and after the price changes. Because prices change only due to policy changes, our OLS results match 2SLS estimates that use the policy changes as instruments. Table 4 summarizes the results. We consider specifications that pool the two states and specifications that treat them separately. As we interpret the latter as providing the better summary of the empirical variation, we use an average of the subsample coefficients in the empirical model. The coefficients are those in columns (iii) and (iv).

The effects of the rate reductions are of broader interest, so in the remainder of this appendix we examine how four variables respond: the number of calls per inmate-month, the minutes per call, the calling minutes per person/day, and the expenditure per inmate-month. Each variable is computed as the average across all incarcerated individuals in the state prison system and is observed monthly. A visual inspection of the data indicates that each variable is reasonably stable before the price changes (Figure D.3), which also supports the validity of our demand estimation. Afterward, the number of calls and calling minutes increases, yet expenditures decrease. In New York, the minutes per call decreased slightly, consistent with incarcerated individuals substituting to shorter calls in response to the elimination of connection charges. In New Jersey, where connection charges did not exist, minutes per call changed much less.

Table D.2 summarizes the changes quantitatively, using "before" and "after" periods that we select based on our visual inspection of the data. The average number of calls per inmate-month increases from 8.82 to 15.86 in New York and from 8.32 to 27.00 in New Jersey. Sample mean tests indicate that these changes differ from zero at the 1% level. Similar patterns are observed for the average minutes per inmate-day spent on the phone. Average expenditures per inmate-month decrease from \$23.77 to \$14.35 in New York and from \$30.36 to \$13.24 in New Jersey, and these changes are again statistically significant. Thus, the raw data support the hypothesis that price is a meaningful determinant of phone usage.

[34]In both states, the same prices were charged for local, intrastate, and interstate calls, and for different payment methods (e.g., debit and collect). In New Jersey, the first change, reducing the per-minute price from \$0.33 to \$0.19, occurred in February 2014 at the direction of the DOC. We observe subsequent reductions to \$0.17 in March 2014, \$0.15 in September 2014, and \$0.044 in May 2015. The last change followed a new state law. New Jersey had no connection charge before or after these changes.

D Additional Figures and Tables

4.5.1 After determining that a proposal satisfies the mandatory requirements, the evaluator(s) shall use both objective and subjective judgement in conducting a comparative assessment of the proposal in accordance with the evaluation criteria stated below:

Call Expense to Called Parties and Inmates	50%
Experience & Reliability	20%
Proposed method of Performance	30%

(a) Scoring Rule from the RFP

				For DPMM Use Only	
NAME OF OFFEROR	Experience And Reliability (Max. 20 Pts.)	Proposed Method of Performance (Max. 30 Pts.)	TOTAL SUBJECTIVE POINTS (Max. 50 Pts.)	COST POINTS INSERTED BY PURCHASING (Max. 50 Pts.)	TOTAL POINTS (Max. 100 Pts.)
1. MCI Worldcom	20	30	= 50 +	50	TOTAL = 100
2. Public Communications Service	15	25	= 40 +	47.75	TOTAL = 87.75

(b) Scoring Sheet for Submitted Bids

Application #1 - Inmate Telephone System

001	$ 0.00	firm, fixed price per minute, local calls;
002	$ 1.30	firm, fixed price per call, set-up charge, local calls
003	$ .14	firm, fixed price per minute, intralata calls;
004	$ 1.00	firm, fixed price per call, set-up charge, intralata calls
005	$ .14	firm, fixed price per minute, interlata calls;
006	$ 1.00	firm, fixed price per call, set-up charge, interlata calls
007	$ .45	firm, fixed price per minute, interstate calls;
008	$ 2.45	firm, fixed price per call, set-up charge, interstate calls

(c) Finalized Prices in the Contract

Figure D.1: Illustrative Example of a Procurement Process

Notes: The figure displays screenshots illustrating the procurement process in Missouri in 2000. On March 7, the Department of Corrections held a "Pre-Proposal Conference" for prospective bidders. A formal RFP was issued on March 29; it detailed the scoring criteria and was made available to all prospective bidders (top panel). Proposals were due April 12. From April 20 to July 18, a committee evaluated the bids and assigned scores to MCI Worldcom and Public Communications Service, the two bidders (middle panel). MCI received the highest score and was awarded the contract at their proposed rates (bottom panel).

On-site Cash Transaction Process Fees		
Low End	Upper End	Cost
$0.00	$300.00	$3.00

On-Site or Over The Internet		
Non-Cash Deposit		
Low End	Upper End	Cost
$0.00	$19.99	$2.95
$20.00	$50.00	$4.25
$50.01	$100.00	$5.50
$100.01	$200.00	$8.50
$200.01	$300.00	$11.50

(a) GTL Account Funding Fees

Fees:

Funding Fee:

Amount Deposited	$5.00 - $25.00	$25.01 - $50.00	$50.01 - $100.00
Funding Fee	$1.99	$2.99	$3.99

Note: Refund Fee $2.00 applies when closing an account and requesting a refund of remaining balance

Policies:

Minimum Deposit Amount: $5
Maximum Deposit Amount: $100

(b) Securus Account Funding Fees

Figure D.2: Illustrative Examples of Ancillary Fee Schedules

Notes: Some bids in our data contain information about the ancillary fees set by providers. The figure shows the ancillary fees of Securus and GTL as described in bids submitted to New Jersey in 2014 (and as captured with screenshots). GTL proposed fees that users would pay when depositing money into an account, and Securus proposed fees that users would pay when closing an account or requesting refunds. CenturyLink also submitted a bid in the 2014 New Jersey auction, but it did not include ancillary fees.

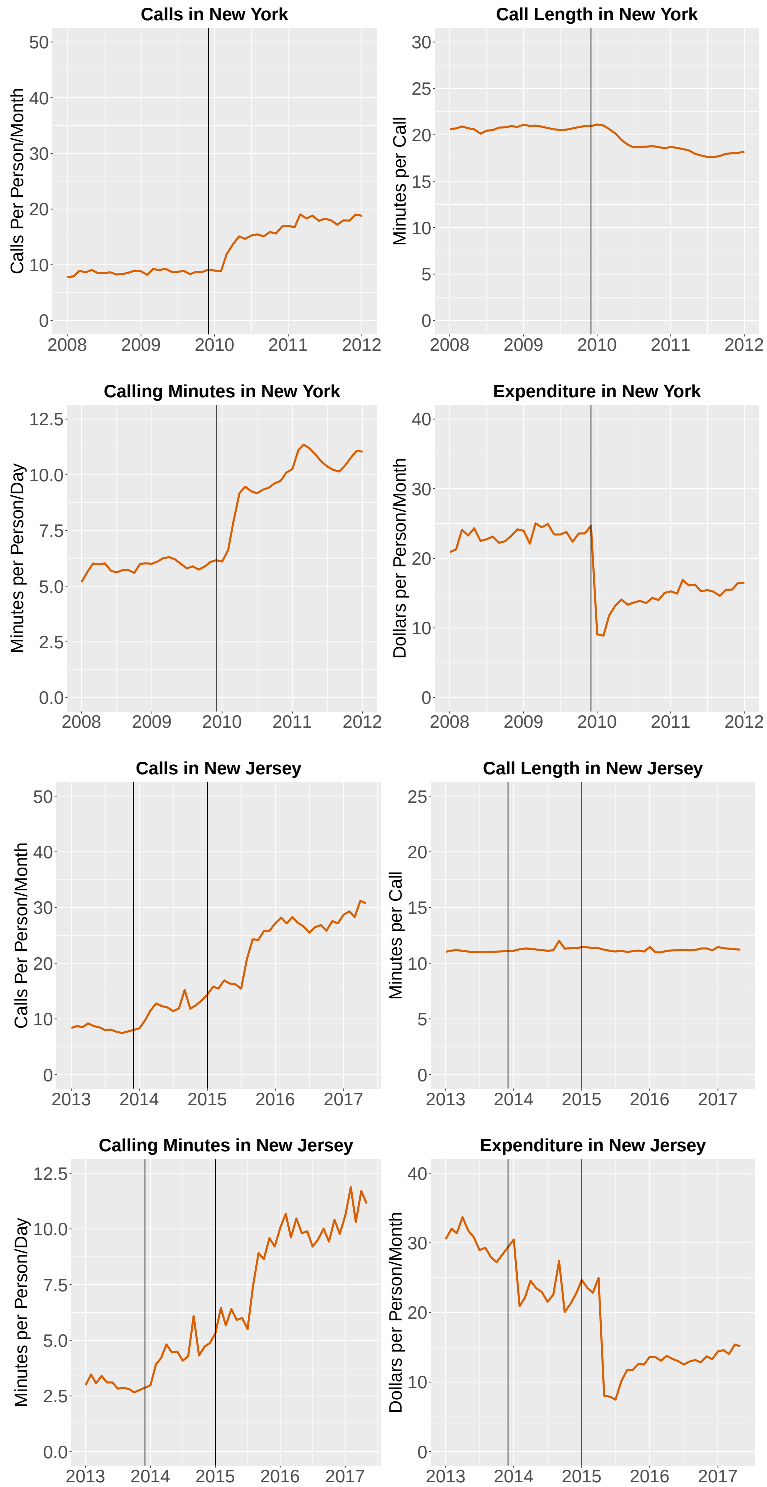

Figure D.3: Calling Patterns Before and After Price Reductions

Notes: The figure plots calls per inmate-month, minutes per call, minutes per person/day, and expenditure per inmate-month over time in New York (top four panels) and New Jersey (bottom four panels). The data points are monthly averages. The vertical black lines show the timing of price changes.

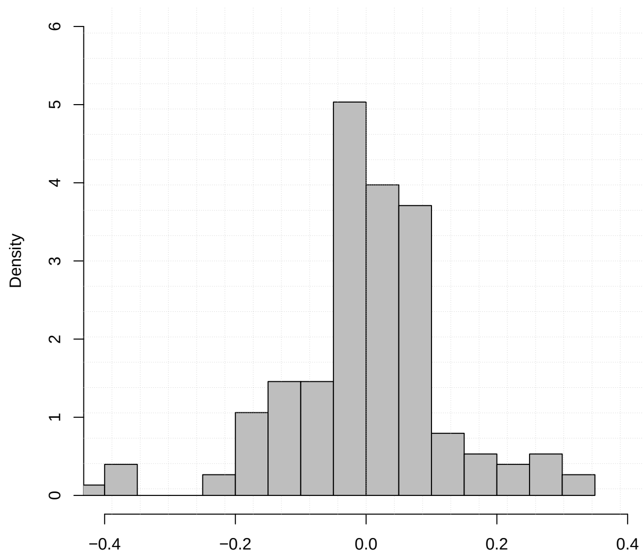

Figure D.4: Distribution of Technical Score Residuals

Notes: The figure provides a histogram of the residuals we obtain from a regression of technical scores on auction fixed effects. We measure the technical score as the score of the bidder divided by the maximum possible score (for all elements of the bid related to technical capabilities or subjective assessments of the provider).

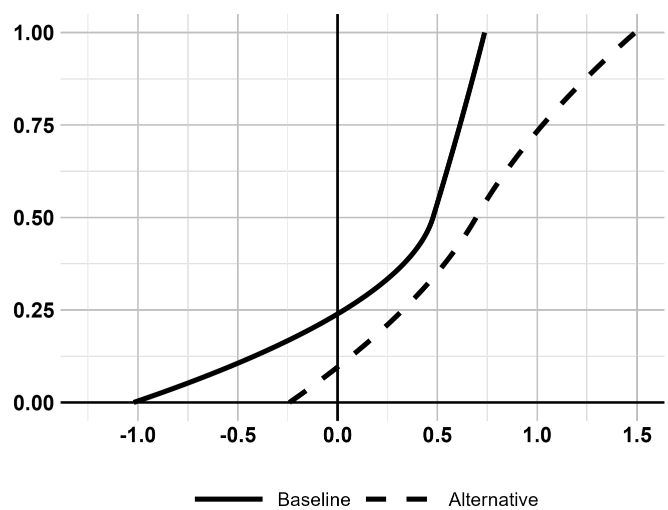

Figure D.5: Sensitivity of Estimated Cost Distribution to Greater Demand

Notes: The figure plots the estimated CDFs of cost (net of fees) obtained with the baseline demand function and an alternative version in which the demand intercept is shifted to produce a one-dollar increase in willingness-to-pay.

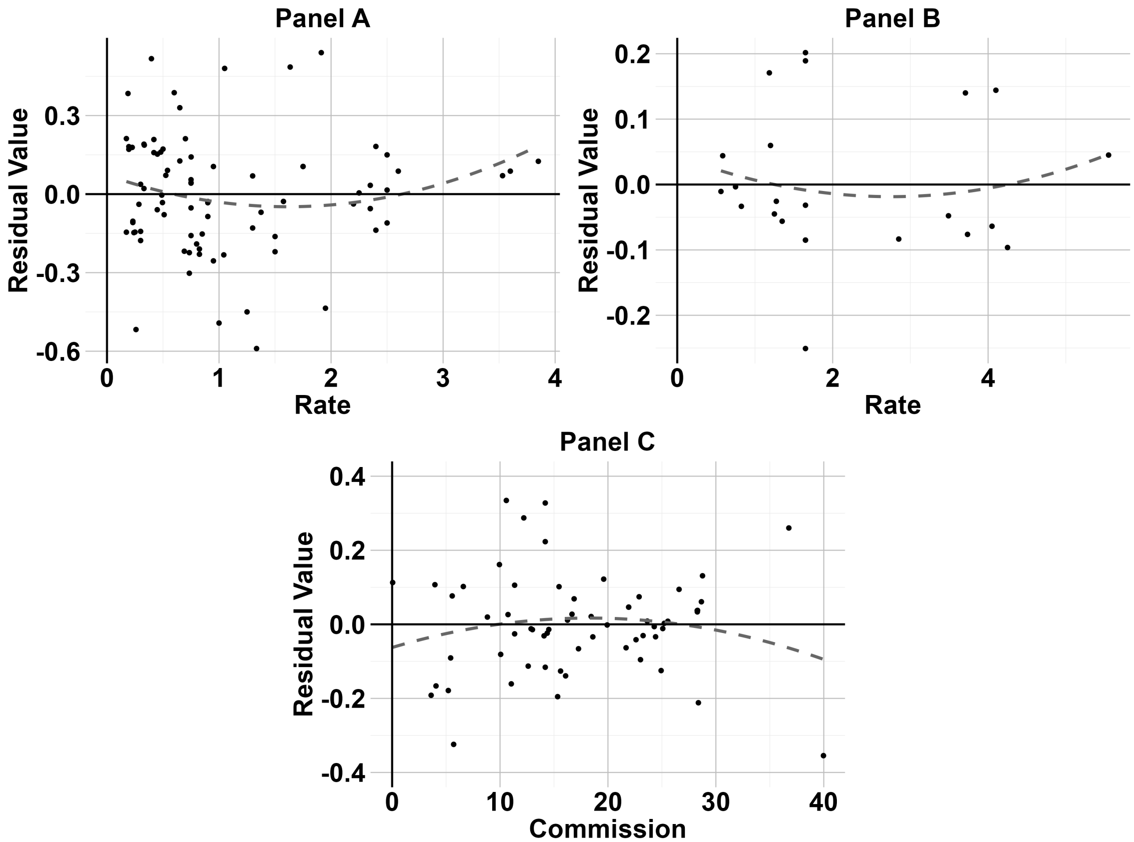

Figure D.6: Assessing Linearity of the Contribution Functions

Notes: The figure plots the residuals obtained from columns (ii), (iii), and (iv) of Table 3 against the rate (Panels A and B) and the commission (Panel C). The dashed gray lines provide quadratic fits. The coefficients on the squared terms (not shown) are small relative to the slope coefficients of Table 3. For Panels A and C, but not Panel B, they are statistically different from zero.

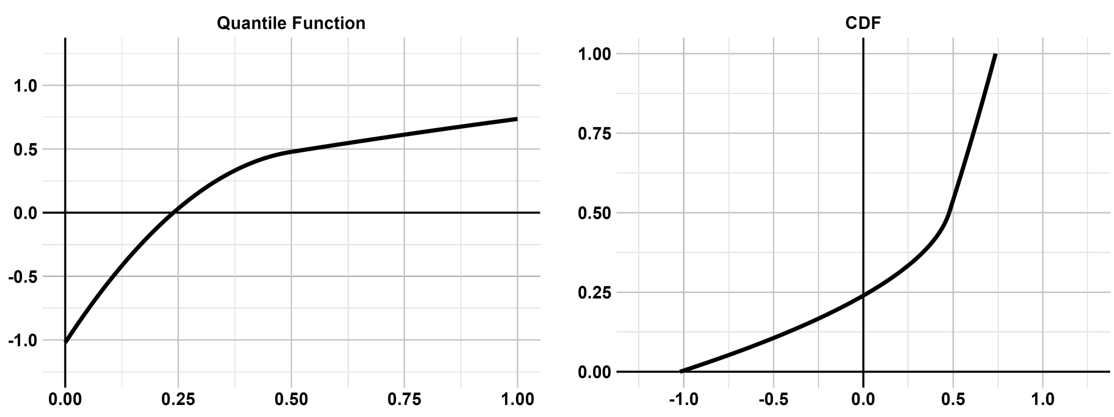

Figure D.7: The Quantile Function and the Implied CDF

Notes: We estimate the parameters of an I-spline that satisfies the requirements to be a *quantile function*, or inverse CDF. We then invert the quantile function to obtain the implied CDF. The left panel shows the quantile function that we estimate with the baseline specification, and the right panel shows the corresponding CDF (see also Figure 2).

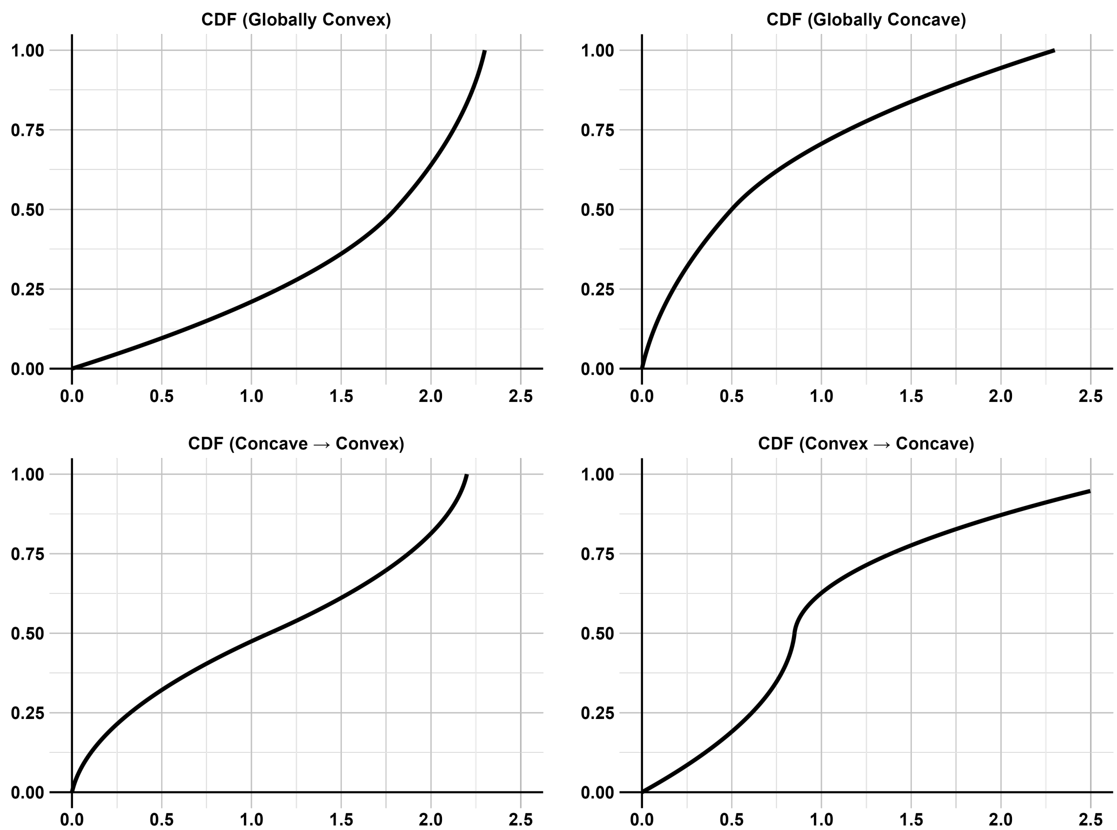

Figure D.8: Illustrating the Flexibility of the I-Spline for CDFs

Notes: The figure illustrates the flexibility of the I-spline in our baseline specification. We set the constant to zero for each panel ($\tau_0 = 0$). In the top left, the CDF is globally convex because the spline weights decrease magnitude ($\tau_1 = 1.4, \tau_2 = 0.8, \tau_3 = 0.1$). In the top right, the CDF is globally concave because the spline weights increase in magnitude ($\tau_1 = 0.1, \tau_2 = 0.8, \tau_3 = 1.4$). In the bottom left, the CDF is locally concave then locally convex, which obtains with a large middle weight ($\tau_1 = 0.1, \tau_2 = 2.0, \tau_3 = 0.1$). In the bottom right, the CDF is locally convex then locally concave, which obtains with a small middle weight ($\tau_1 = 0.8, \tau_2 = 0.1, \tau_3 = 2.0$).

Table D.1: Summary of the Bid Data

	State	Year	Rate Weight	Technical Weight	Comm. Weight	# of Bids	# of Bidders	Winning Bidder	Contract Rate	Contract Commission Percentage	Contract Commission Fixed	Contract Commission Total
1	Alaska	2016	0	0.56	0.44	4	4	Securus	1.05	0.94	0	19.61
2	Arizona	2014	0	0.17	0.83	4	4	CenturyLink	1.84	0.94	0	28.27
3	Arkansas	2014	0	1	0	4	4	Securus	4.80	0.73	0	11.65
4	Florida	2013	0.05	0.90	0.05	3	3	GTL	0.75	0.67	0	10.58
5	Florida	2017	0.17	0.83	0	3	3	Securus	0.19	0	0	0
6	Georgia	2015	0	0.65	0.35	3	3	Securus	1.95	0.97	6.73	36.76
7	Idaho	2014	0.20	0.80	0	5	5	CenturyLink	2.40	0	20.00	20.00
8	Illinois	2012	0.45	0	0.55	3	3	Securus	4.10	0.87	0	22.89
9	Indiana	2010	0.22	0.67	0.11	3	3	PCS	4.05	0.40	0	10.74
10	Kentucky	2012	0	0.74	0.30	5	5	GTL	2.25	0.88	0	28.66
11	Kentucky	2017	0	0.86	0.14	3	3	GTL	1.65	0.50	0	14.19
12	Maine	2015	0.15	0.70	0.15	4	4	Legacy	1.65	0.55	0	15.61
13	Massachusetts	2013	0	0.86	0.14	6	4	GTL	2.36	0.60	0	19.93
14	Michigan	2018	0	1	0	3	3	GTL	2.40	0	23.65	23.65
15	Minnesota	2005	0	0.84	0.16	4	4	MCI	1.75	0.49	0	14.37
16	Minnesota	2016	0.40	0.60	0	8	4	GTL	0.33	0.47	0	3.60
17	Minnesota	2019	0.30	0.70	0	4	4	GTL	0.33	0.40	0	3.04
18	Missouri	2000	0.75	0.25	0	2	2	MCI	1.30	0	0	0
19	Missouri	2006	0.52	0.48	0	5	5	PCS	2.50	0	0	0
20	Missouri	2011	0.42	0.58	0	10	7	Securus	1.75	0	0	0
21	Missouri	2018	0.28	0.72	0	3	3	Securus	0.75	0	0	0
22	Montana	2017	0.20	0.80	0	5	5	CenturyLink	0.54	0	9.24	9.24
23	Nebraska	2008	0.33	0.67	0	6	6	PCS	0.70	0	0	0
24	Nebraska	2016	0.41	0.59	0	6	5	GTL	0.19	0	0	0
25	New Hampshire	2013	1	0	0	4	4	IC Solutions	0.65	0.20	9.67	12.48
26	New Hampshire	2018	0.35	0.65	0	4	4	GTL	0.19	0.20	11.12	12.04
27	New Jersey	2014	0.40	0.60	0	2	2	GTL	0.40	0	0	0
28	North Dakota	2016	0.23	0.68	0.09	5	5	Securus	1.19	0.25	0	5.70
29	Oklahoma	2018	0.30	0.70	0	4	4	Securus	1.91	0	14.57	14.57
30	Utah	2014	0	0.70	0.30	4	4	CenturyLink	2.10	0.90	0	28.76
31	Utah	2019	0	0.70	0.30	4	4	GTL	1.80	0.95	0	28.28
32	Vermont	2010	0	0.70	0.30	5	5	PCS	2.30	0.37	0	12.20
33	Vermont	2016	0.03	0.95	0.02	3	3	GTL	0.58	0.41	0	5.21
34	Virginia	2005	0.20	0.70	0.10	2	2	MCI	5.55	0.41	0	0.05
35	West Virginia	2014	0.30	0.70	0	3	3	CenturyLink	0.48	0	0	0.01
36	Wisconsin	2008	0.30	0.70	0	6	6	Embarq	1.05	0.30	0	6.25
37	Wisconsin	2018	0.20	0.80	0	3	3	CenturyLink	0.17	0.30	0	1.23

Notes: The table summarizes the auction-level data. Rate is the cost of a 15-minute local collect phone call. The commission percentage is the percentage of the non-fee revenue that the provider pays to the state. The fixed commission is a fixed amount of money that the provider pays to the state, converted to be in dollars per inmate-month. The total commission combines these two forms of payments and is in dollars per inmate-month.

Table D.2: Calling Patterns Before and After Price Reductions

	New York				New Jersey			
	Before	After	Change	p-value	Before	After	Change	p-value
Number of Calls	8.82	15.86	7.04	0.000	8.32	27.00	18.68	0.000
Minutes per Call	20.81	18.87	-1.94	0.000	11.05	11.18	0.13	0.011
Minutes per Day	6.04	9.83	3.80	0.000	3.03	9.91	6.87	0.000
Expenditures	23.77	14.35	-9.43	0.000	30.36	13.24	-17.12	0.000

Notes: The table provides the average number of calls per inmate-month, average minutes per call, average minutes per inmate-day, and average expenditure per inmate-month (in dollars), both before and after price reductions in New York and New Jersey. For each, it also provides the change and the p-value from a sample means test of the null hypothesis that the shift equals zero. We observe the variables at the month-state level; the p-values are computed assuming independence. For New York, we use a "before" period of January-December 2009 and an "after" period of April 2010 - March 2011. For New Jersey, we use a "before" period of January-October 2013, as some data are unavailable for November-December 2013, and an "after" period of January-December 2016.

Table D.3: Regressions with Linear Belief Functions

State	Year	Constant: $\gamma_0^{(21)}$	Cost: $\gamma_1^{(21)}$	Tech Score: $\gamma_2^{(21)}$	R^2	MSE
Florida	2017	0.8077	-0.0256	0.5485	0.9404	0.0002
Idaho	2014	0.7246	-0.0436	0.4523	0.7785	0.0014
Minnesota	2016	0.7457	-0.0597	0.3517	0.9590	0.0001
Minnesota	2019	0.6004	-0.0497	0.4743	0.9689	0.0002
Missouri	2000	1.0718	-0.0517	0.0978	0.9624	0.0000
Missouri	2006	0.8889	-0.0785	0.3086	0.9857	0.0000
Missouri	2011	0.9671	-0.0645	0.3955	0.9793	0.0001
Missouri	2018	0.9535	-0.0305	0.3673	0.9276	0.0003
Montana	2017	0.7886	-0.0358	0.5143	0.9518	0.0001
Nebraska	2008	0.6847	-0.0582	0.4985	0.9768	0.0001
Nebraska	2016	0.7416	-0.0666	0.4101	0.9759	0.0001
New Hampshire	2018	0.7746	-0.0707	0.4048	0.7733	0.0045
New Jersey	2014	0.5702	-0.0469	0.3894	0.8351	0.0008
Oklahoma	2018	0.6823	-0.0563	0.3929	0.7845	0.0034
Wisconsin	2008	0.7423	-0.0512	0.4415	0.9429	0.0000
Wisconsin	2018	0.8197	-0.0299	0.5124	0.9490	0.0002
West Virginia	2014	0.9390	-0.0346	0.3827	0.9452	0.0001

Notes: For auctions in which the commission does not enter the scoring rule, we approximate symmetric Bayes-Nash equilibrium by computing how 100 "focal firms" would bid, given a belief that competitors' bids are a linear function of their costs and technical score. We regress the best responses of the focal firms (e.g., their score bid) on their costs and technical scores, update beliefs, and iterate. The table summarizes the regression results for each auction after 21 iterations. We provide the coefficients, the R^2 of each state-specific regression, and the MSE between the final belief and regression coefficients.

Table D.4: Determinants of Belief Coefficients with Linear Beliefs

	Dependent Variable		
	$\gamma_1^{(21)}$	$\gamma_2^{(21)}$	$\gamma_0^{(21)}$
Constant	-0.0004	-0.0749	0.6728
	(0.0102)	(0.0538)	(0.1357)
Rate Weight	-0.0712		0.4002
	(0.0174)		(0.2311)
Technical Weight		0.6424	
		(0.0724)	
# of Bidders	-0.0061	0.0148	-0.0038
	(0.0017)	(0.0072)	(0.0229)
R^2	0.6494	0.8642	0.1837

Notes: The table summarizes how the belief coefficients correlate with selected auction features. The belief coefficients $(\gamma_1, \gamma_2, \gamma_0)$ correspond to the coefficient on the cost, the technical score, and the constant in the belief regressions (reported in Table D.3). Here, we regress the belief coefficients on the rate weight, the technical weight, and the number of bidders. The unit of observation is an auction, and there are 17 observations. The technical weight equals one less the rate weight, as the commission does not enter the scoring rule for these auctions. Each regression also includes a constant. Standard errors are in parentheses.

Table D.5: Regressions with Flexible Belief Functions

State	Year	Constant	Cost	Tech Score	Cost^2	Tech Score^2	Interaction	R^2	MSE
Florida	2017	0.8113	-0.0269	0.5759	-0.0055	-1.5707	0.1480	0.9784	0.0025
Idaho	2014	0.7129	-0.0413	0.5416	-0.0008	0.4932	-0.0038	0.8497	0.0000
Minnesota	2016	0.7426	-0.0578	0.3180	-0.0090	-0.2359	0.0942	0.9727	0.0000
Minnesota	2019	0.5973	-0.0453	0.4058	-0.0071	-0.3775	0.1359	0.9870	0.0006
Missouri	2000	1.0731	-0.0552	0.1074	-0.0060	-0.3430	0.0507	0.9472	0.0001
Missouri	2006	0.8916	-0.0793	0.2851	-0.0057	-0.4566	0.1125	0.9924	0.0000
Missouri	2011	0.9778	-0.0721	0.3935	-0.0104	-0.5837	0.1443	0.9924	0.0000
Missouri	2018	0.9661	-0.0434	0.4520	-0.0086	-1.0989	0.2393	0.9810	0.0000
Montana	2017	0.7866	-0.0332	0.4714	-0.0046	-0.1696	0.0592	0.9451	0.0000
Nebraska	2008	0.6857	-0.0569	0.4489	-0.0091	-0.6400	0.1634	0.9921	0.0000
Nebraska	2016	0.7433	-0.0667	0.3622	-0.0106	-0.4887	0.1626	0.9913	0.0015
New Hampshire	2018	0.7816	-0.0691	0.5128	-0.0175	-1.4608	0.3261	0.9084	0.0002
New Jersey	2014	0.5720	-0.0564	0.4040	-0.0271	-2.5715	0.4301	0.8341	0.0006
Oklahoma	2018	0.6832	-0.0616	0.5409	-0.0165	-2.1807	0.3247	0.9783	0.0075
Wisconsin	2008	0.7358	-0.0517	0.4142	-0.0054	0.2753	0.0823	0.9501	0.0000
Wisconsin	2018	0.8226	-0.0317	0.5338	-0.0048	-1.4429	0.1584	0.9795	0.0001
West Virginia	2014	0.9497	-0.0451	0.4334	-0.0102	-1.0233	0.2303	0.9826	0.0000

Notes: For auctions in which the commission does not enter the scoring rule, we approximate symmetric Bayes-Nash equilibrium by computing how 100 "focal firms" would bid, given a belief that competitors' bids are a second-order polynomial in their costs and technical score. We regress the focal firms' best responses (e.g., their score bid) on the corresponding variables, update beliefs, and iterate. The table summarizes the regression results for each auction after 42 iterations (21 with the linear beliefs, 21 with the flexible belief function). We provide the coefficients, the R^2 of each state-specific regression, and the MSE between the final belief and regression coefficients.